

Affidavit by Certified Professional Pursuant to OAC 3745-300-13(P) and 3745-300-05(F)(4)

[for submittals under OAC 3745-300-13(P): Certified Professionals submitting to Ohio EPA an addendum to a No Further Action Letter.]

Volunteer:

Port of Greater Cincinnati Development Authority

Owner(s) of Property:

Hamilton County Regional Business Park LLC
7870 East Kemper Road, Suite 240
Cincinnati, Ohio 45249

Chronis Properties LTD.
320 South Anthony Wayne Avenue
Lockland, Ohio 45215

Certified Professional who issued the NFA Letter:

Ihor Melnyk
KERAMIDA Inc.
260 Northland Boulevard, Suite 114
Cincinnati, Ohio 45246

Property subject to NFA Letter:

Hamilton County Regional Business Park
320 South Anthony Wayne Avenue
Lockland, Hamilton County, Ohio 45215

State of Ohio)

County of Hamilton)

ss:

I, Ihor Melnyk, being first duly sworn according to law, state that, to the best of my knowledge, information and belief:

1. I am an adult over the age of eighteen (18) years old and competent to testify herein.
2. I am a Certified Professional, No. 147, in good standing under Ohio Revised Code (ORC) Chapter 3746 and Ohio Administrative Code (OAC) Chapter 3745-300.
3. I prepared a No Further Action (NFA) Letter, that I submitted on or about February 16, 2009 on behalf of Port Authority of Greater Cincinnati, for the Hamilton County regional Business Park LLC property located at 320 South Anthony Wayne Avenue, Lockland, Hamilton County, Ohio
4. I am submitting with this affidavit an addendum to the NFA Letter.
5. The purpose of this addendum is to provide additional supporting documentation for the NFA letter as requested in the Notice of Deficiencies and Reviewer Comments letter from the Ohio EPA dated June 23, 2009.
6. This addendum includes the following information, data, documents, or reports:

Document

NFA Letter Addendum and Response to Comments

Date of Document

August 19, 2009

7. I have read the standards of conduct contained in OAC 3745-300-05(F), and maintained full compliance with these standards regarding the NFA Letter, as amended by the addendum, and while rendering professional services to the Volunteer regarding the Property.

8. The Property is eligible for the Voluntary Action Program pursuant to ORC 3746.02 and OAC 3745-300-02.
9. The voluntary action has been conducted and the NFA Letter, as amended by the addendum, has been issued in accordance with the ORC Chapter 3746 and OAC Chapter 3745-300.
10. The Property meets the applicable standards contained in ORC Chapter 3746 and OAC Chapter 3745-300.
11. The voluntary action at the Property was conducted in compliance with all applicable federal, state and local laws and regulations.
12. The NFA Letter, as amended by the addendum, the completed NFA Form for the Property and any other information, data, documents and reports submitted with the NFA Letter and the NFA Form are true, accurate and complete.
13. The NFA Letter, as amended by the addendum, the completed NFA Form and all supporting information, data, documents and reports, are a true, accurate and complete characterization of conditions at the Property, including the presence or absence of hazardous substances and petroleum.

Further affiant sayeth naught.



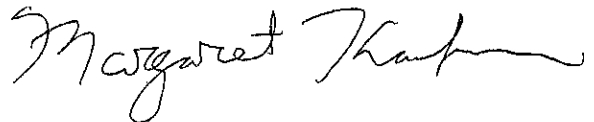
Signature of Affiant

Sworn to before me this 19th day of August, 2009



MARGARET KAUFMAN
Notary Public, State of Ohio
My Commission Expires 11-27-2012

Notary Public



Revised 10/02



KERAMIDA

A Global EHS Services Provider

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Cincinnati, Ohio 45246
(513) 769-9057 • Fax (513) 772-5727
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August 19, 2009

Mr. Joseph M. Smindak
Senior Site Coordinator
Division of Emergency and Remedial Response
Ohio EPA
401 East Fifth Street
Dayton, Ohio 45402-2911

Re: NFA Letter Addendum and Response to Comments
Former Celotex Property
320 South Anthony Wayne Avenue, Lockland, Hamilton County, Ohio 45215
KERAMIDA Project 2-11708

Dear Mr. Smindak:

KERAMIDA received your comment letter dated June 23, 2009 in regard to the No Further Action (NFA) letter prepared for the above-referenced property on behalf of the Port of Greater Cincinnati Development Authority. This addendum was prepared to provide a response to the comments and includes the documentation that was requested. A copy of the comment letter is provided for reference as Attachment 1. The responses are provided in the same order as the comment letter.

1. Quality of Data Tables

The data tables (Tables 2, 3, 4, and 6) of the BHE Phase II report provided in the NFA letter were prepared by another consultant (BHE), and were only available as a scan of a paper copy. The original reports, including the tables, had not been saved by BHE. As such, the tables have been re-created and are included as Attachment 2 to this addendum letter. The tables have been thoroughly reviewed to ensure accuracy. Copies of the laboratory reports were included in the Phase II report and are available to check the data.

2. Background Demonstration

The mention of the soil results being consistent with the Ohio background levels in Section 3.2 of the Risk Assessment was only a statement to put the results in context. It was not intended to be used as any sort of background demonstration or imply that the Property soils do not require further evaluation. The metals were not eliminated as constituents of potential concern (COPCs) and were included in the risk assessment.

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3. Ground Water Data

The ground water data Table 6 in the Risk Assessment differs from Table 2 of the BHE Phase II because Table 6 documents only the constituents of concern (COCs) detected in ground water at the Property. The data in Table 2 of the BHE Phase II report document that no other potential COCs were detected and that the detection limits were at or below the Voluntary Action Program (VAP) standards. The Certified Professional (CP) double-checked the detection limits against the VAP standards and confirmed that all of the detection limits were below the VAP standards. Therefore, the Risk Assessor did not see a need for Table 6 of the Risk Assessment to include COCs that were not detected in any ground water samples. A note stating the above has been added to the table and a copy of the revised table is provided to this addendum as Attachment 3.

4. Risk Mitigation Plan

The Risk Mitigation plan has been revised to include the information requested to meet requirements of OAC 3745-300-15. It includes a legal description and a survey drawing of the Property with the area covered by the plan clearly depicted. The revised Risk Mitigation Plan is provided as Attachment 4 to this addendum.

5. Incorrect Classification of Ground Water Zone 2

As stated in the comment letter, ground water zone 2 does not require classification because of the Protection of Ground Water Meeting Unrestricted Potable Use Standards (POGWMUPUS) demonstration. Zone 2 was never classified in either the BHE or Petro Phase II reports. BHE identified one saturated zone at the site (which was later shown to be the lower aquifer-zone 2) as a Critical Resource because it yields over 100 gpm. Petro identified two saturated zones, one that produces over 100 gpm, and an upper zone classified as Class B based on yield. Petro did not classify Zone 2. In preparation of the NFA Checklist, it appeared that the form required a classification to be checked. The CP completed part 5 of NFA checklist G with the box checked indicating the lower zone is Class A. In discussion with Mr. Joe Smindak and Mr. Mike Proffitt of the Ohio EPA on August 17, 2009 it was agreed that the CP should add a check box and note to the NFA checklist stating that classification is not required because of the POGWMUPUS demonstration. A revised copy of NFA checklist section G is provided as Attachment 5.

6. Demonstration of Compliance with Applicable Standards for Ground Water Zone 1

This comment is similar to comment #1. The tables have been re-created and are included as Attachment 2 to this addendum letter.

7. Ground Water Classification

Section G, item 1.h of the NFA checklist incorrectly referenced section 5.6 of the BHE Phase II report as the location of the evaluation of ground water use in the area. The information necessary for the evaluation was provided in Appendix 7 of the Petro Phase II Property Assessment Addendum dated May 2005. This includes the ODNR well log (9921394) for the Phillip Carey Manufacturing Company (a.k.a Celotex). This water was used for manufacturing purposes and was abandoned during remediation of the property. A well sealing report was forwarded to ODNR on July 30, 2008. Ground water is no longer used and a ground water use restriction via the environmental covenant has been developed. A revised copy of section G of the NFA Checklist is provided as Attachment 5.

8. Ground Water Yield

The ground water yield test data used to classify the ground water have been reviewed and determined to meet the VAP requirements. Specific demonstrations that the yield tests were performed in accordance with the VAP requirements are as follows:

- VAP Technical Guidance Compendium (TGC) VA30007.09.013 states that yield tests should be completed during March through May to bias the results to the typical time of year with the highest yield. The test was performed on May 3, 2005 which is during the required period. The date of the test was documented in the Petro Phase II report. In addition, it rained on six of the 13 days prior to the yield test, and snowed on 2 others. Thus, the yield test was biased to a time expected to result in the highest yield.
- TGC VA30007.09.014 requires the testing to be completed in the area of the Property likely to have the highest yield and recommends testing areas where soil boring data indicate the saturated zone is thickest. The locations of the wells used for the test had the largest amount of screened interval in the saturated zone. The two wells tested had 5 and 9.5 feet of screen in saturated zone. The other wells averaged 3.25 feet.
- TGC VA30007.09.015 requires the use statistics to determine a representative yield. Because the yields of the two wells varied, Petro used the maximum yield instead of calculating a representative yield (more conservative).
- TGC VA30007.09.016 requires adjustment of the measured yield for well size and percent penetration of the saturated zone thickness. The yield test calculations in the Petro Phase II report document these adjustments. The adjusted yield using the maximum yield was 2.2 gallons per minute, which is a typical yield for Hamilton County glacial till.

9. Ground Water Sampling Procedures

The ground water sampling procedures used are consistent with industry standards and many past Ohio VAP projects. As stated in OAC 3745-300-07(E)(3), sampling may be performed in accordance with either the Ohio EPA "Technical Guidance Manual for Hydrogeologic Investigations and Ground Water Monitoring" (TGM) or:

- (a) Documents containing data collection, field testing and sampling techniques which are demonstrated to:
 - (i) Be field-validated;
 - (ii) Be documented and peer-reviewed;
 - (iii) Ensure the representativeness of samples taken following the technique; and
 - (iv) Be proven capable of achieving the data quality needs identified in paragraph (E) of this rule; or

The COCs at the Property included volatile organic compounds (VOCs) and polynuclear aromatic compounds (PAHs). The concern regarding the use of bailers is the loss of VOCs in the sample. The CP evaluated the data in the BHE report relative to Unrestricted Potable Use Standard (UPUS) and confirmed that the concentrations of VOCs and the detection limits for other VOCs were well below the VAP UPUS. The CP calculated the ratio of the detected concentrations to UPUS and determined that the benzene concentration detected in well MW-6 had the highest ratio of 0.26 ($1.3 \mu\text{g/l}$ detected / $5 \mu\text{g/l}$ UPUS). Thus, up to 74% of the benzene could have been lost to volatilization during sampling before the result would reach the $5 \mu\text{g/l}$ UPUS. It is unlikely that sampling using the bailers would have resulted in a loss of greater than 74% of the benzene or any other VOC. Use of bailers would not be expected to affect PAH concentrations.

The CP researched technical journals for information regarding validity of samples collected from wells using bailers. Volume 14, Issue 4 of "Ground Water Monitoring & Remediation" published by the National Ground Water Association contains an article titled "Field Comparison of Micropurging vs. Traditional Ground Water Sampling." The article discusses the methods and results of the sampling and states that there was no significant difference between the two data sets within a 95% confidence interval using the nonparametric sign test. A copy of the article is provided as Attachment 6.

Because naphthalene is considered a VOC for vapor intrusion and it was detected in ground water, soil gas sampling was performed to evaluate the pathway. No naphthalene was detected in the soil gas.

A ground water use restriction is being placed on the Property. Thus, deviations in dissolved COCs would not have a bearing on the Property conditions as they relate to a No Further Action status.

Relative to the comment regarding use of a surge block to develop the wells, the CP agrees that it is not a preferred means of well development for fine-grained soils. However, the development method would not be expected to result in a decrease in the concentrations of COCs in the ground water.

10. Ground Water Response Requirements

The statement regarding the ground water response requirement for ground water zone 2 was prepared by another consultant who incorrectly classified the ground water. As stated in subsequent documents, no response is required for zone 2.

11. VAP Eligibility

The NFA letter documented the acceptance of the hazardous waste generator closure report by the Ohio EPA. A copy of the report was included in Volume III of the NFA Letter in Appendix 3 of the Phase II Property Assessment-Addendum 2. However, a copy of the report is also provided as Attachment 7 to this addendum.

We trust that the responses to the comments, including the attached documents, will be satisfactory to the Ohio EPA to grant a Covenant Not to Sue for the Property. We appreciate your timely review of the NFA letter and look forward to completing this project. Please feel free to call me if you have any further comments regarding the NFA letter or this document.

Sincerely,
KERAMIDA Inc.



Ihor Melnyk, P.E.
Project Manager

Attachments

c: Mr. Keith Egan
Ms. Christine Russell



Certified Professional Seal

IHOR MELNYK
CERTIFIED PROFESSIONAL (CP147)
ORC Section 3746.04(B)(5)
OAC Rule 3745-300-03
My certification expires 08-30-2009

Attachment 1

Comment Letter



State of Ohio Environmental Protection Agency

Southwest District Office

401 E. Fifth St.
Dayton, Ohio 45402

TELE: (937) 285-6357 FAX: (937) 285-8249
www.epa.state.oh.us

Ted Strickland, Governor
Lee Fisher, Lieutenant Governor
Chris Korleski, Director

CERTIFIED MAIL

June 23, 2009

Ihor Melnyk, CP147
Keramida, Inc.
260 Northland Boulevard, Suite 114
Cincinnati, Ohio 45246

RE: Ohio EPA Review of the Voluntary Action Program (VAP) No Further Action (NFA) Letter for Hamilton County Regional Business Park LLC 09NFA334; Notice of Deficiencies and Reviewer Comments

Dear Mr. Melnyk:

On February 25, 2009, Ohio EPA Southwest District Office received the NFA Letter for the property located at 320 South Wayne Avenue, Lockland, Hamilton County (the "Property"). You submitted the NFA Letter on behalf of the Port of Greater Cincinnati Development Authority (the Volunteer) with an accompanying request for a covenant not to sue under the Voluntary Action Program, Ohio Revised Code (ORC) Chapter 3746 and Ohio Administrative Code (OAC) Chapter 3745-300. All comments developed by Ohio EPA personnel in a collective review of the NFA Letter are presented below.

1. QUALITY OF DATA TABLES

Tables included in the BHE Phase II contained sections that were shaded, making the numbers in those shaded-in sections illegible. The data tables submitted must be legible for review by VAP staff. Please resubmit legible versions of tables 2, 3, 4 and 6 of the BHE Phase II.

2. BACKGROUND DEMONSTRATION- OAC 3745-300-07(H)

Section 3.2 of the Risk Assessment Documents states that for Identified Area 3 soil, each detected metal was retained as a COPC for the screening level assessment. However, on page 3 of the Risk Assessment Documents, the table labeled Phase II Results lists COCs within the IAs, and states that for metals levels were consistent with Ohio background levels; however, no background demonstration was submitted. If metals were eliminated as COPC from IAs using the background argument, then a background demonstration must be completed pursuant to OAC 3745-300-07(H).

3. GROUNDWATER DATA

The groundwater COCs presented in Table 6 of the Risk Assessment Documents are different than the groundwater COCs presented in Table 2 of the BHE Phase II Property Assessment. Please explain the discrepancies between the groundwater data sets.

4. RISK MITIGATION PLAN- OAC 3745-300-15(G)

A. Legal Description of the Portion of the Property Subject to the Risk Mitigation Plan- OAC 3745-300-11(F)(1)(f)

Pursuant to OAC 3745-300-11(F)(1)(f), if, based on the results of the property-specific risk assessment, the entire property is not subject to the RMP, a legal description of that portion of the property subject to the RMP must be attached to and incorporated by reference in the RMP. According to Section 1.0 of the RMP, the purpose of the RMP is to mitigate potential exposure and health risks related to subsurface soil in Identified Area 3 (IA-3) at the Property; therefore, the entire Property is not subject to the RMP. Please attach a legal description of that portion of the property subject to the RMP, and incorporate that by reference into the RMP.

B. Handling Environmental Media at the Property- OAC 3745-300-11(F)(1)(d)

Pursuant to OAC 3745-300-11(F)(1)(d), the RMP must contain directions on how to handle environmental media at the property that may contain chemicals of concern. Please clarify how environmental media at the property that may contain chemicals of concern will be handled.

C. Precautions Communicated to Employees- OAC 3745-300-11(F)(1)(i)

Pursuant to OAC 3745-300-11(F)(1)(i), the RMP must contain a summary explanation of the precautions that each contractor must require of and communicate to its employees and subcontractors.

Please include a summary explanation of the precautions that each contractor must require of and communicate to its employees and subcontractors in the revised version of the RMP.

5. The Volunteer incorrectly determined that Zone 2 is classified as a Class A resource.

Section 7.0 of the Petro Phase II Property Assessment Addendum states that saturated Zone 2 is capable of providing ground water yields between 800 and 1400 GPM. As defined in OAC 3745-300-10(B)(1), any saturated zone capable of yielding over 100 GPM is a Critical Resource. However, since protection of ground water that meets unrestricted potable use standards (POGWMUPUS) demonstration was made for this zone, it does not need to be classified.

The Volunteer should remove language referring to the classification of Zone 2 as a Class A resource.

6. OAC 3745-300-07(G)(1) Demonstration of compliance with applicable standards. The Volunteer needs to demonstrate compliance with applicable standards for Ground Water in Zone 1 as directed in OAC 3745-300-07(G)(1).

OAC 3745-300-07(G)(1) requires that, "The data collected in accordance with paragraphs (D) and (E) of OAC 3745-300-07 must be sufficient to determine whether applicable standards are met in all identified areas and affected media".

Section G, item 1.g of the Phase II Property Assessment checklist states that the ground water sampling results can be found in Tables 2 & 3 of the NFA Phase II, however, the tables were illegible.

Legible tables must be prepared and submitted to Ohio EPA.

7. OAC 3745-300-07(D)(8). Ground Water Classification. In order to properly classify ground water, the volunteer must identify any visual evidence of ground water use and must review ODNR water well log information for the properties on which ground water exceeds UPUS, as required in OAC 3745-300-07(D)(8)(a)(i) through (ii).

According to Section G, item 1.h of the Phase II Property Assessment checklist, this information is included in section 5.6 of the BHE Phase II Property Assessment, however, review showed that section 5.6 only discusses the protection of ground water that meets unrestricted potable use standards (POGWMUPUS) demonstration.

The required information must be collected and provided to Ohio EPA.

8. OAC 3745-300-07(D)(9)(b). Determination of Ground Water Yield. When yield testing is performed for ground water classification, the volunteer

must bias the yield testing locations to the area of highest yield as required by OAC 3745-300-07(D)(9)(b).

Section 7.0 of the Petro Phase II Property Assessment does not include information indicating that the yield testing locations were properly biased.

The Volunteer must provide the information to Ohio EPA.

9. OAC 3745-300-07(E)(3). Ground Water Sampling Methods and Procedures. The volunteer did not follow the methods and procedures described in OAC 3745-300-07(E)(3).

OAC 3745-300-07(E)(3) states that a volunteer must follow the methods and procedures according to their limitations and intended uses contained in the "Technical Guidance Manual For Hydrogeologic Investigations and Ground Water Monitoring (TGM)," (Ohio EPA February 1995), except when the application the document would be inconsistent with the purpose of the "Phase II Property Assessment" and this chapter.

According to the BHE Phase II Property Assessment, the Volunteer used bailers and surge blocks to develop the monitoring wells installed at the property, and used bailers to collect VOC samples from shallow monitoring wells.

Page 10-10, paragraph 1 of the TGM states that when collecting ground water samples, "The use of bailers is discouraged." The TGM goes on to state in paragraph 2 that though bailers are discouraged, they are appropriate if other technologies cannot be used. This is not the case at Celotex, especially in light of the fact that variable speed pumps were used to sample some monitoring wells at the property.

Additionally, page 8-8 of the TGM states that surging of wells in fine grained sediments should be avoided.

The Volunteer must demonstrate to Ohio EPA that these deviations from the TGM do not significantly impact the usability of ground water data for this NFA.

10. OAC 3745-300-10(F) Ground Water Response Requirements. The volunteer incorrectly determined that the appropriate Ground Water Response for Saturated Zone 2 is OAC 3745-300-10(F)(6)(a).

Section 5.6 of the BHE Phase II Property Assessment is the documentation of the POGWMUPUS demonstration for Saturated Zone 2. Since Zone 2 meets unrestricted potable use standards (UPUS), there is no ground water response for the zone.

11.OAC 3745-300-02(C)(5)(b) VAP Eligibility of Property where closure of a hazardous waste facility is required.

Generator closure activities associated with a loading dock area where containerized hazardous wastes were temporarily stored represent a potential eligibility issue for participation in the Voluntary Action Program. However, by letter dated May 4, 2009, from Harold O'Connell of Ohio EPA's Division of Hazardous Waste Management, to Keith Egan of Petro Environmental, LLC, this eligibility issue was resolved. Please provide a copy of the approved Generator Closure Report.

Response Required

The foregoing provides Ohio EPA concerns and identified deficiencies regarding the NFA Letter demonstration of compliance with applicable standards at the Property. The matters must be addressed and the NFA Letter deficiencies corrected before Ohio EPA can issue a covenant not to sue (CNS) to the Volunteer for the Property. To allow the NFA Letter review to proceed, as Certified Professional, you can issue an addendum to the NFA Letter that includes the needed corrections to the NFA Letter and checklist and other documentation.

The NFA Letter addendum would need to be prepared so that the NFA Letter, as amended, meets the criteria of OAC 3745-300-13. Further, the addendum needs to be submitted under the Certified Professional affidavit provision of OAC 3745-300-13(P), and include any other necessary affidavits (such as Volunteer or Certified Laboratory affidavits). Please submit two copies of the addendum to my attention, as well as another copy and the original copy of the addendum to the attention of Records Management Officer, DERR, Ohio EPA Central Office, within sixty (60) days of your receipt of this letter. This letter does not constitute an action of the Director and is not subject to appeal.

If you have any questions related to this review or would like to schedule a meeting or conference call, please feel free to contact me at (937) 285-6064 or by e-mail at joe.smindak@epa.state.oh.us.

Sincerely,



Joseph M. Smindak
Senior Site Coordinator
Division of Emergency and Remedial Response

cc: VAP File Manager, DERR/CO

JMS/plh

Attachment 2

Data Tables

Table 2. Groundwater Sample VOC Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	Water Samples (in ug/L, parts per billion)					Generic Unrestricted Potable Use Standards (ug/L)
	MW4, W	MW5, W	MW6, W	MW7, W	MW8, W	
Acetone	<20	<20	<20	<20	<20	1,600
Benzene	<1	<1	<1	<1	<1	5
tert-Butylbenzene	1.2	1.2	<1	<1	<1	NE
sec-Butylbenzene	11.9	11.3	<1	<1	<1	NE
n-Butylbenzene	3	2.8	<1	<1	<1	NE
Bromochloromethane	<1	<1	<1	<1	<1	NE
Bromodichloromethane	<1	<1	<1	<1	<1	NE
Bromoform	<1	<1	<1	<1	<1	NE
Bromobenzene	<1	<1	<1	<1	>1	NE
2-Butanol (MEK)	<12.5	<12.5	<12.5	<12.5	<12.5	6,800
Carbon disulfide	<1	<1	<1	<1	<1	1,400
Carbon tetrachloride	<1	<1	<1	<1	<1	5
Chlorobenzene	<1	<1	<1	<1	<1	100
Chloroethane	<5	<5	<5	<5	<5	6,100
2-Chlorotoluene	<1	<1	<1	<1	<1	NE
4-Chlorotoluene	<1	<1	<1	<1	<1	NE
Chloroform	<1	<1	<1	<1	<1	50
Chloromethane	<5	<5	<5	<5	<5	5
Dibromochloromethane	<1	<1	<1	<1	<1	19
Dibromomethane	<1	<1	<1	<1	<1	NE
Dichlorodifluoromethane	<1	<1	<1	<1	<1	1,900
1,2-Dibromo-3-chloropropane	<5	<5	<5	<5	<5	NE
1,2-Dichlorobenzene	<1	<1	<1	<1	<1	600
1,3-Dichlorobenzene	<1	<1	<1	<1	<1	13
1,4-Dichlorobenzene	<1	<1	<1	<1	<1	75
1,1-Dichloroethane	<1	<1	<1	<1	<1	1,400
1,2-Dichloroethane	<1	<1	<1	<1	<1	5
1,1-Dichloroethene	<1	<1	<1	<1	<1	7
cis-1,2-Dichloroethene	<1	<1	<1	<1	<1	70
trans-1,2-Dichloroethene	<1	<1	<1	<1	<1	100
1,2-Dichloropropane	<1	<1	<1	<1	<1	5
1,3-Dichloropropane	<1	<1	<1	<1	<1	NE
2,2-Dichloropropane	<1	<1	<1	<1	<1	NE
1,1-Dichloropropene	<1	<1	<1	<1	<1	NE
cis-1,3-Dichloropropene	<1	<1	<1	<1	<1	14
trans-1,3-Dichloropropene	<1	<1	<1	<1	<1	NE
Ethyl benzene	3.3	<1	1.4	<1	<1	700
Hexachlorobutadiene	<5	<5	<5	<5	<5	NE
n-Hexane	<5	<5	<5	<5	<5	560
2-Hexanone	<12.5	<12.5	<12.5	<12.5	<12.5	NE
Isopropylbenzene (Cumene)	6.3	5.6	<1	<1	<1	1,300
Isopropyltoluene	<1	4.8	<1	<1	<1	NE
Bromomethane	<5	<5	<5	<5	<5	NE
Methylene chloride	<5	<5	<5	<5	<5	NE
4-Methyl-2-pentanone (MIBK)	<12.5	<12.5	<12.5	<12.5	<12.5	760
n-Propylbenzene	<1	<1	<1	<1	<1	NE
Styrene	<1	<1	<1	<1	<1	100
1,1,1,2-Tetrachloroethane	<1	<1	<1	<1	<1	44
1,1,1,2,2-Tetrachloroethane	<1	<1	<1	<1	<1	5.9
Tetrachloroethene	<1	<1	<1	<1	<1	5
Toluene	<1	<1	1.0	<1	<1	1,000
1,2,4-Trichlorobenzene	<5	<5	<5	<5	<5	70
1,1,1-Trichloroethane	<1	<1	<1	<1	<1	200
1,1,2-Trichloroethane	<1	<1	<1	<1	<1	5
Trichloroethene	<1	<1	<1	<1	<1	5
Trichlorofluoromethane	<1	<1	<1	<1	<1	3,700
1,2,3-Trichloropropane	<5	<5	<5	<5	<5	NE
1,2,4-Trimethylbenzene	14.8	4.8	1.5	<1	<1	290
1,3,5-Trimethylbenzene	5.1	1.7	<1	<1	<1	300
Vinyl Acetate	<5	<5	<5	<5	<5	8,400
Vinyl chloride	<1	<1	<1	<1	<1	2
Xylenes	10.8	2.7	3.7	<1	<1	10,000

Notes:

"<" indicates that the result is less than the indicated reporting limit.

NE = no standard has been established for this compound.

Shading indicates samples collected from the carved-out area of IA-1.

Table 3. Soil and Groundwater BTEX Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	Soil samples (results in ug/kg, ppb)																		VAP Generic Standards (in ppb)		
	GB-9, 3'	GB-19, 3.5'	GB-11,4'	GB-11, 12'	GB-12, 3'	GB-13, 3'	GB-13,12'	GB-16,3'	GB-17,3'	GB-17, 12'	GB-18, 3.5'	GB-19, 3'	GB-20, 3'	GB-21, 3'	GB-22, 3'	GB-23, 3'	MW4, 3.5'	MW4, 23.5'	Industrial Direct Contact	Construction Direct Contact	Leach-based
Benzene	<13	<13	<13	<12	<11	<13	<11	<12	<12	<11	<13	<14	<11	<13	<12	<12	<6.2	<51	100,000	310,000	17
Ethylbenzene	<13	<13	<13	<12	<11	<13	<11	<12	<12	<11	<13	<14	<11	<13	27	<12	<6.2	<51	230,000	230,000	12,000
Toluene	<13	<13	<13	<12	<11	<13	<11	<12	<12	<11	<13	<14	<11	<13	24	<12	<6.2	<51	520,000	520,000	6,800
Xylenes	<13	<13	<13	<12	<11	<13	<11	<12	<12	<11	<13	<14	<11	<13	156	<12	<6.2	<51	160,000	160,000	160,000

Parameter	Water Samples (in ug/L, parts per billion)								VAP Generic Unrestricted Pot. Use Standard (ug/L)
	MW1, W	MW2, W	MW3, W	MW4, W	MW5, W	MW6, W	MW7, W	MW8, W	
Benzene	<1	<1	<1	<1	<1	1.3	<1	<1	5
Ethylbenzene	<1	<1	<1	3.3	<1	1.4	<1	<1	700
Toluene	<1	<1	<1	<1	<1	1.0	<1	<1	1,000
Xylenes	<1	<1	<1	10.8	2.7	3.7	<1	<1	10,000

Notes:

1. "<" Indicates that the result is less than the indicated reporting limit.

2. Generic direct-contact standards are for industrial use and apply to soil within 2 feet of the surface.

3. Leach-based standards apply to soils between two feet of surface and groundwater. The generic leach-based standards for soil type 1 are listed.

4. Shading Indicates samples collected from the carved-out area of 1A-1

Table 4. Soil and Groundwater Sample SVOC Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	Soil Sample Results (ug/kg, ppb)																		
	GB-2, 3.5'	GB-3, 3'	GB-3, 12'	GB-4, 3'	GB-5, 3.5'	GB-5, 12'	GB-6, 3'	GB-9, 3'	GB-10, 3.5'	GB-11, 4'	GB-11, 12'	GB-12, 3'	GB-13, 3'	GB-13, 12'	GB-16, 3'	GB-17, 3'	GB-17, 12'	GB-18, 3.5'	GB-19, 3'
Acenaphthene	<8,200	<7,500	1,000,000	<395	<395	<386	<4,000	489	<432	<419	<386	502	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Acenaphthylene	<8,200	<7,500	420,000	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Anthracene	<8,200	<7,500	1,940,000	<395	<395	<386	<4,000	<419	<432	<419	<386	581	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzo(a)anthracene	32,800	<7,500	2,210,000	<395	975	<386	<4,000	<419	<432	577	<386	1,680	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzo(b)fluoranthene	41,000	<7,500	2,240,000	<395	2,000	<386	<4,000	<419	<432	888	<386	1,400	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzo(k)fluoranthene	23,100	<7,500	852,000	<395	929	<386	<4,000	<419	<432	<419	<386	760	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzo(a)pyrene	31,700	<3,700	1,750,000	<198	1,560	<193	<2,000	<209	<216	613	<193	1,100	<216	<183	<20,200	<3,800	<187	<4,400	<4,600
Benzo(g,h,i)perylene	13,900	<7,500	569,000	<395	638	<386	<4,000	<419	<432	<419	<386	1,240	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzyl alcohol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Butyl benzyl phthalate	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Bis(2-Chloroethyl)ether	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Bis(2-Chloroethoxy)methane	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Bis(2-Ethylhexyl)phthalate	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	1,160	<386	1,640	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2,2'-Oxybis(1-chloropropane)	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
4-Bromophenyl-phenylether	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
4-Chloroaniline	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2-Chloronaphthalene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Chrysene	20,700	<7,500	1,940,000	<395	854	<386	<4,000	<419	<432	524	<386	1,170	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Dibenz(a,h)anthracene	5,250	<3,700	173,000	<198	230	<193	<2,000	<209	<216	<209	<193	253	<216	<183	<20,200	<3,800	<187	<4,400	<4,600
Dibenzofuran	<8,200	<7,500	2,450,000	<395	<395	<386	<4,000	590	<432	<419	<386	2,430	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
1,2-Dichlorobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
1,3-Dichlorobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
1,4-Dichlorobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
3,3'-Dichlorobenzidine	<16,000	<15,000	<99,400	<790	<789	<772	<8,100	<838	<864	<838	<773	<738	<865	<730	<81,000	<15,000	<747	<18,000	<18,000
Diethylphthalate	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Dimethyl phthalate	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2,4-Dinitrotoluene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2,6-Dinitrotoluene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Di-n-octylphthalate	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Fluoranthene	30,300	<7,500	<4,970,000	<395	1,130	<386	<4,000	584	<432	816	<386	1,360	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Fluorene	<8,200	<7,500	3,340,000	<395	<395	<386	<4,000	448	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Hexachlorobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Hexachloro-1,3-butadiene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Hexachlorocyclopentadiene	<16,000	<15,000	<99,400	<790	<789	<722	<8,100	<838	<864	<838	<773	<738	<865	<730	<81,000	<15,000	<747	<18,000	<18,000
Hexachloroethane	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Indeno(1,2,3-cd)pyrene	12,400	<7,500	617,000	<395	667	<386	<4,000	<419	<432	<419	<386	795	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Isophorone	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Naphthalene	<8,200	<7,500	4,220,000	<395	<395	<386	<4,000	8,250	<432	<419	<386	8,230	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Nitrobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
N-Nitroso-di-n-propylamine	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Phenanthrene	<8,200	<7,500	<4,970,000	<395	667	<386	<4,000	1,020	<432	862	<386	7,350	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Pyrene	<82,000	8,380	<4,970,000	<395	2,070	<386	4,430	508	<432	2,610	<386	<3,690	<433	<365	<40,500	10,800	<374	<8,900	14,200
1,2,4-Trichlorobenzene	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Benzoic acid	<41,000	<37,000	<248,000	<1,980	<1,970	<1,930	<20,000	<2,090	<2,160	<2,090	<1,930	<1,850	<2,160	<1,830	<202,000	<38,000	<1,870	<44,000	<46,000
4-Chloro-3-methylphenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2-Chlorophenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2,4-Dichlorophenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2,4-Dimethylphenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	1,030	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2-Methyl-4,6-dinitrophenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2-Methylphenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	454	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
meta & para-Methylphenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	379	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
2-Nitrophenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200
Pentachlorophenol	<8,200	<7,500	<49,700	<395	<395	<386	<4,000	<419	<432	<419	<386	<369	<433	<365	<40,500	<7,700	<374	<8,900	<9,200

Table 4. Soil and Groundwater Sample SVOC Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	GB-20, 3'	GB-21, 3'	GB-22, 3.5'	GB-23, 3'	MW2, 3'	MW2, 23.5'	MW4, 3.5'	MW4, 23.5'	Industrial Direct Contact	Construction Direct Contact	Leach-based
Acenaphthene	<3,600	<416	17,100	<384	<38,300	<362	<4,100	<401	180,000,000	30,000,000	NE
Acenaphthylene	<3,600	<416	8,950	<384	<38,300	<362	<4,100	<401	18,000,000	3,000,000	NE
Anthracene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	880,000,000	100,000,000	NE
Benzo(a)anthracene	5,590	<416	33,100	<384	<38,300	<362	<4,100	<401	63,000	810,000	NE
Benzo(b)fluoranthene	8,380	<416	135,000	<384	<38,300	<362	<4,100	<401	63,000	810,000	NE
Benzo(k)fluoranthene	<3,600	<416	42,900	<384	<38,300	<362	<4,100	<401	630,000	81,000,000	NE
Benzo(a)pyrene	6,220	<208	126,000	<192	<19,100	<181	<2,000	<200	6,300	81,000	NE
Benzo(g,h,i)perylene	<3,600	<416	51,800	<384	<38,300	<362	<4,100	<401	9,100,000	NE	NE
Benzyl alcohol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Butyl benzyl phthalate	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	220,000	NE	NE
Bis(2-Chloroethyl)ether	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Bis(2-Chloroethoxy)methane	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Bis(2-Ethylhexyl)phthalate	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	230,000	230,000	NE
2,2'-Oxybis(1-chloropropane)	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
4-Bromophenyl-phenylether	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
4-Chloroaniline	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
2-Chloronaphthalene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Chrysene	5,200	<416	27,700	<384	<38,300	<362	<4,100	<401	6,700,000	11,000,000	NE
Dibenz(a,h)anthracene	<1,800	<208	17,900	<192	<19,100	<181	<2,000	<200	6,700	41,000	NE
Dibenzofuran	<3,600	<416	11,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
1,2-Dichlorobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	370,000	170,000	NE
1,3-Dichlorobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	240,000	240,000	NE
1,4-Dichlorobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	470,000	5,800,000	NE
3,3'-Dichlorobenzidine	<7,300	<832	<18,000	<768	<76,600	<724	<8,200	<802	450,000	1,200,000	NE
Diethylphthalate	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	640,000	640,000	NE
Dimethyl phthalate	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	2,800,000	NE	NE
2,4-Dinitrotoluene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	5,800,000	1,700,000	NE
2,6-Dinitrotoluene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	2,900,000	3,800,000	NE
Di-n-octylphthalate	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	10,000,000	10,000,000	NE
Fluoranthene	10,800	<416	37,200	<384	<38,300	<362	<4,100	<401	33,000,000	170,000,000	NE
Fluorene	<3,600	<416	16,700	<384	<38,300	<362	<4,100	<401	120,000,000	346,000,000	NE
Hexachlorobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	120,000	370,000	NE
Hexachloro-1,3-butadiene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	580,000	170,000	NE
Hexachlorocyclopentadiene	<7,300	<832	<18,000	<768	<76,600	<724	<8,200	<802	NE	NE	NE
Hexachloroethane	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	2,900,000	3,600,000	NE
Indeno(1,2,3-cd)pyrene	<3,600	<416	47,000	<384	<38,300	<362	<4,100	<401	67,000	110,000	NE
Isophorone	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	4,600,000	4,600,000	NE
Naphthalene	5,790	<416	12,700	1,790	<38,300	<362	<4,100	<401	530,000	1,800,000	2,300
Nitrobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	370,000	1,700,000	NE
N-Nitroso-di-n-propylamine	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Phenanthrene	6,570	<416	28,600	<384	<38,300	<362	<4,100	<401	91,000	NE	NE
Pyrene	15,400	<416	182,000	<384	<38,300	<362	<4,100	<401	25,000,000	130,900,000	NE
1,2,4-Trichlorobenzene	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Benzoic acid	<18,000	<2,080	<44,000	<1,920	<191,100	<1,810	<20,000	<2000	NE	NE	NE
4-Chloro-3-methylphenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
2-Chlorophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
2,4-Dichlorophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
2,4-Dimethylphenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	59,000,000	180,000,000	NE
2-Methyl-4,6-dinitrophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
2-Methylphenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	15,000,000	4,300,000	NE
meta & para-Methylphenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	150,000/8,500 ppm	430,000/4,300 ppm	NE
2-Nitrophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	NE	NE	NE
Pentachlorophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	240,000	1,700,000	NE
Phenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	1,000,000,000	510,000,000	2,200
2,4,5-Trichlorophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	290,000,000	860,000,000	NE
2,4,6-Trichlorophenol	<3,600	<416	<8,800	<384	<38,300	<362	<4,100	<401	18,000,000	54,000,000	NE

Notes:

1. "<" Indicates that the result is less than the Indicated reporting limit
2. Generic direct-contact standards are for commercial land use and apply to soil within 2 feet of the surface.
3. Leach-based standards apply to soils between two feet of surface and groundwater. The generic leach-based standards for soil type I are listed.
4. NE = no standard has been established for this compound.
5. VAP Unrestricted Potable Use standard applies to groundwater.
6. * - indicates that the listed standard is from "Supplemental Generic Numerical Values: VAP Technical Assistance", and not from OAC 3745-300-08.

Results in bold text exceed a VAP standard.

Results in italic text indicate that the analytical reporting limit is greater than a VAP generic standard.

Shading indicates sample results from the carvedout area of IA-1.

Table 4. Soil and Groundwater Sample SVOC Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	Water Sample Results (ug/L, ppb)							Resample		Unrestricted Potable Use Standard
	MW1, W	MW2, W	MW3, W	MW4, W	MW5, W	MW6, W	MW7, W	MW8, W	MW8, W	
Acenaphthene	<1	<1	<1	21.7	2,290	23,300	<1	<1	<1	680
Acenaphthylene	<1	<1	<1	55.1	4,800	49,800	<1	<1	<1	660*
Anthracene	<1	<1	<1	1.59	1,300	15,500	<1	<1	<1	2,600
Benzo(a)anthracene	<0.2	<0.2	<0.2	0.36	550	5,450	<0.2	0.42	0.27	0.29*
Benzo(b)fluoranthene	<0.2	<0.2	<0.2	0.79	262	2,780	<0.2	0.26	<0.2	0.21*
Benzo(k)fluoranthene	<0.2	<0.2	<0.2	<0.2	214	2,100	<0.2	<0.2	<0.2	1.8*
Benzo(a)pyrene	<0.2	<0.2	<0.2	<2.0	443	4,640	<0.2	0.43	0.31	0.2
Benzo(g,h,i)perylene	<0.2	<0.2	<0.2	<0.2	93.1	3,790	<0.2	0.44	<0.2	63
Benzyl alcohol										NE
Butyl benzyl phthalate										2,900
Bis(2-Chloroethyl)ether										NE
Bis(2-Chloroethoxy)methane										NE
Bis(2-Ethylhexyl)phthalate										19
2,2'-Oxybis(1-chloropropane)										NE
4-Bromophenyl-phenylether										NE
4-Chloroaniline										NE
2-Chloronaphthalene										NE
Chrysene	<0.2	<0.2	<0.2	0.21	409	4,730	<0.2	0.43	0.22	47
Dibenz(a,h)anthracene	<0.2	<0.2	<0.2	<0.2	41.1	395	<0.2	<0.2	<0.2	0.00097*
Dibenzofuran										NE
1,2-Dichlorobenzene										NE
1,3-Dichlorobenzene										13
1,4-Dichlorobenzene										NE
3,3'-Dichlorobenzidine										NE
Diethylphthalate										13,000
Dimethyl phthalate										12,000*
2,4-Dinitrotoluene										32
2,6-Dinitrotoluene										16
Di-n-octylphthalate										6
Fluoranthene	<0.2	<0.2	<0.2	1.42	2,200	24,900	<0.2	1.03	0.78	370
Fluorene	<1	<1	<1	19.6	3,100	35,200	<1	<1	<1	500
Hexachlorobenzene										1
Hexachloro-1,3-butadiene										NE
Hexachlorocyclopentadiene										50
Hexachloroethane										15
Indeno(1,2,3-cd)pyrene	<0.2	<0.2	<0.2	<0.2	230	2,440	<0.2	0.31	0.22	0.14*
Isophorone										1,700
Naphthalene	<2	<2	<2	300	23,000	270,000	<2	<2	<2	140
Nitrobenzene										NE
N-Nitroso-di-n-propylamine										NE
Phenanthrene	<1	<1	<1	9.77	4,560	51,800	<1	<1	<1	2,600*
Pyrene	<0.2	<0.2	<0.2	1.5	1,600	17,900	<0.2	0.94	0.69	280*
1,2,4-Trichlorobenzene										290*
Benzoic acid										NE
4-Chloro-3-methylphenol										NE
2-Chlorophenol										NE
2,4-Dichlorophenol										NE
2,4-Dimethylphenol										310
2-Methyl-4,6-dinitrophenol										NE
2-Methylphenol										780
meta & para-Methylphenol										780 / 78
2-Nitrophenol										NE
Pentachlorophenol										1
Phenol										9,400
2,4,5-Trichlorophenol										1,400
2,4,6-Trichlorophenol										120

Notes:

1. "<" indicates that the result is less than the indicated reporting limit
2. Generic direct-contact standards are for commercial land use and apply to soil within 2 feet of the surface.
3. Leach-based standards apply to soils between two feet of surface and groundwater. The generic leach-based standards for soil type I are listed.
4. NE = no standard has been established for this compound.
5. vAP Unrestricted Potable Use standard applies to groundwater.
6. * - Indicates that the listed standard is from "Supplemental Generic Numerical Values: VAP Technical Assistance", and not from OAC 3745-300-08.

Results in bold text exceed a VAP standard.

Results in italic text indicate that the analytical reporting limit is greater than a VAP generic standard.

Shading indicates sample results from the carvedout area of IA-1.

Table 6. Soil Sample TPH Analytical Results
Former Celotex Facility
Lockland, Ohio

Parameter	Soil Samples (Results in mg/kg, ppm)																		VAP TPH Soil Saturation
	GB-9, 3'	GB-10, 3.5'	GB-11, 4'	GB-11, 12'	GB-12, 3'	GB-13, 3'	GB-13, 12'	GB-16, 3'	GB-17, 3'	GB-17, 12'	GB-18, 3.5'	GB-19, 3'	GB-20, 3'	GB-21, 3'	GB-22, 3.5'	GB-23, 3'	MW4, 3'	MW4, 23.5'	Concentration (in ppm)
Light Fraction	848	<27	<27	<27	<23	<28	<23	<26	<24	<24	953	33	<23	<26	<28	<24	<26	1,430	5,000
Middle fraction	83.5	<13	<63	<12	519	<13	<11	<123	67.2	<11	3,020	4,850	<11	<13	3,120	24.2	64.2	22.5	10,000
Heavy fraction	131	<65	708	<59	633	<66	<55	3,030	865	<57	9,140	25,600	1,310	<63	11,700	113	239	<24	20,000

Notes:
"<" indicates that the result is less than the indicated reporting limit
TPH standards are based on silty/clayey sand (Kv = 10 E-04 to 10 E-05).
Results in bold text exceed a VAP standard.
Shading indicates samples collected from the carved-out area of IA-1.

Attachment 3

Revised Table 6 of Risk Assessment Report

Table 6
Identified Area 7 Property-Wide Groundwater Organics Analytical Results (ug/L)
Former Celotex Property
Lockland, Ohio
KERAMIDA Project No. 2-11708

Well No.	Sample Depth (feet bgs)	Benzene	Toluene	Ethylbenzene	Total Xylenes	Acenaphthene	Acenaphthylene	Anthracene	Benzo(a) anthracene	Benzo(b) fluoranthene	Benzo(k) fluoranthene	Benzo(a) pyrene	Benzo(g,h,i) perylene	Chrysene	Dibenz(a,h) anthracene	Fluoranthene	Fluorene	Indeno(1,2,3-cd) pyrene	Naphthalene	Phenanthrene	Pyrene
MW-1	18-28	<1	<1	<1	<1	<1	<1	<1	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<1	<0.2	<2	<1	<0.2
MW-2	16-26	<1	<1	<1	<1	<1	<1	<1	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<1	<0.2	<2	<1	<0.2
MW-3	17-27	<1	<1	<1	<1	<1	<1	<1	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<1	<0.2	<2	<1	<0.2
MW-4	19-29	<1	<1	3.3	10.8	2.2	5.5	1.59	0.36	0.79	<0.2	<0.2	<0.2	210	<0.2	1.4	19.8	<0.2	300	9.8	1.5
MW-5	Unknown	<1	<1	<1	2.7	2,300	4,800	1,300	550	262	214	93	93	409	41.1	2,200	3,100	230	23,200	4,560	1,600
MW-6	Unknown	1.3	1	1.4	3.7	23,300	49,800	15,600	5,450	2,780	2,100	3,790	3,790	4,730	396	24,900	35,200	2,440	270,000	51,600	17,900
MW-7	34-44	<1	<1	<1	<1	<1	<1	<1	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<1	<0.2	<2	<1	<0.2
MW-8	23-33	<1	<1	<1	<1	<1	<1	<1	0.42	0.26	<0.2	0.43	0.44	0.43	<0.2	1.03	<1	0.31	<2	<1	0.94
Generic Unrestricted Potable Use Standards ⁽¹⁾		5	1,000	700	10,000	680	NS	2,600	NS	NS	NS	0.2	NS	47	NS	370	500	NS	140	NS	280
Generic Unrestricted Potable Use Standards ⁽²⁾		NS	NS	NS	NS	NS	670	NS	0.26	0.17	1.7	NS	63	NS	0.021	NS	NS	0.22	NS	2,800	NS
Frequency of Detection		1/8	1/8	2/8	3/8	3/8	3/8	3/8	4/8	4/8	2/8	3/8	3/8	4/8	2/8	4/8	3/8	3/8	3/8	3/8	4/8
Minimum Detected Concentration		1.3	1	1.4	2.7	2.2	5.5	1.59	0.36	0.26	214	0.43	0.44	0.43	41.1	1.03	19.8	0.31	300	9.8	0.94
Maximum Detected Concentration		1.3	1	3.3	10.8	23300	49800	15600	5450	2780	2100	3790	3790	4730	396	24900	35200	2440	270000	51600	17900
Mean Detected Concentration		1.3	1	2.35	5.733	8534	18202	5634	1500	760.8	1157	1294	1294	1337	218.6	6776	12773	890.1	97833	18723	4876
Median Detected Concentration		1.3	1	2.35	3.7	2300	4800	1300	275.2	131.4	1157	93	93	309.5	218.6	1101	3100	230	23200	4560	800.8
Standard Deviation of Detections		--	--	1.344	4.416	12839	27470	8655	2646	1352	1334	2162	2162	2268	251	12127	19483	1347	149540	28563	8716

Notes:
(1) Ohio EPA VAP, 10/21/02 single chemical standard Shaded values exceed the generic standard. bgs = below ground surface -- = Not applicable due low frequency of detection < = not detected at listed detection limit
(2) Ohio EPA VAP Supplemental Unrestricted Potable Use Values, Technical Assistance, January 2008. Single Chemical Standard

Data shown in this table represent only the VOCs and SVOCs detected at the Property. VOCs and SVOCs not shown were not detected in any samples at the Property. All detection limits were less than the VAP UPUS.

Risk Management Plan



KERAMIDA

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**RISK MITIGATION PLAN RISK MITIGATION PLAN:
PRECAUTIONS FOR AVOIDING POTENTIAL HEALTH RISKS
DURING EXCAVATION ACTIVITIES**

**320 SOUTH WAYNE AVENUE
LOCKLAND, OHIO
KERAMIDA PROJECT NO. 2-11708**

Submitted to:

PORT AUTHORITY OF GREATER CINCINNATI

Ms. Christine Russell
1014 Vine Street, Suite 1600
Cincinnati, Ohio 45202

Submitted by:

KERAMIDA INC.

260 Northland Boulevard, Suite 114
Cincinnati, Ohio 45246
(513) 769-9057

August 22, 2008

Revised July 17, 2009

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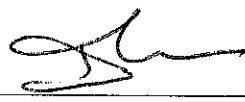
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- 1 Location Map

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**RISK MITIGATION PLAN:
PRECAUTIONS FOR AVOIDING POTENTIAL HEALTH RISKS
DURING EXCAVATION ACTIVITIES**

**320 SOUTH ANTHONY WAYNE AVENUE
LOCKLAND, OHIO
KERAMIDA PROJECT NO. 2-11708**

1.0 PURPOSE

This Risk Mitigation Plan (RMP) has been prepared for the Property located at 320 South Anthony Wayne Avenue, Lockland, Hamilton County, Ohio. The purpose of the RMP is to mitigate potential exposure and health risks related to subsurface soil in Identified Area 3 (IA-3) located in the southeast corner of the Property. The RMP has been prepared in accordance with OAC §3745-300-15(G).

Investigation sampling revealed asbestos in soil in IA-3 and concentrations of polynuclear aromatic hydrocarbons (PAH) in IA-3 soil exceeding the Ohio Environmental Protection Agency (OEPA) Voluntary Action Program (VAP) direct contact standards for construction workers (OEPA 2002). Therefore, the RMP was developed to provide information for use in determining appropriate work practices for subsurface workers to prevent excess exposure and risks from asbestos and PAHs in subsurface soils in IA-3. It is the Property owner's responsibility to implement and maintain this Risk Mitigation Plan. All contractors that perform subsurface construction and excavation activities while on the Property must comply with all OSHA requirements.

Figure 1 shows the Property location. Attachment 1 contains a property survey depicting the location of IA-3 and a legal description of IA-3. The sections below present information on the following:

- Potential health effects associated with exposure to asbestos and PAH
- Precautions to be taken at IA-3 to prevent exposure to asbestos and PAH
- Proper methods of handling asbestos and PAH-impacted soil
- Implementation of the RMP
- Handling of environmental media
- Annual notification to OEPA of RMP status and use
- Criteria for termination of the RMP

2.0 NOTICE TO EMPLOYEES AND CONTRACTORS

It is the Property owner's responsibility to notify employees, contractors or other personnel who have potential to be exposed to contaminated soils in IA-3 regarding the contents of this plan and the requirement for adherence to it. Specifically, the Property owner must provide a copy of the plan and clearly instruct those personnel in the contents and requirements of sections 3.0 through 5.0.

3.0 POTENTIAL HEALTH EFFECTS

The chemicals of concern (COC) at IA-3 are asbestos and PAH in subsurface soil (greater than 2 feet below grade). The potential health effects of these COCs are discussed in the sections that follow.

3.1 ASBESTOS

Asbestos is classified an A1 confirmed human carcinogen for all types of asbestos fibers. Occupationally exposed workers showed increased mortality and incidence of lung cancer, mesotheliomas, and gastrointestinal cancer. Animal studies have shown similar results.

3.2 POLYNUCLEAR AROMATIC HYDROCARBONS

In general, exposure to polynuclear aromatic hydrocarbons (PAHs) can result in the following symptoms (HSDB 2008):

- Eyes - Photosensitivity and irritation
- Lungs - Irritation with cough and bronchitis
- Mouth -- Leukoplakia (white plaques on mucous membranes and tongue)
- Skin - precancerous lesions enhanced by UV light exposure, cysts, nodules, dermal burns, photosensitivity

Specific PAHs present in subsurface soil at IA-3 at concentrations greater than the subsurface worker direct contact criteria are:

- Benzo(a)anthracene
- Benzo(b)fluoranthene

- Benzo(a)pyrene
- Dibenzo(a,h)anthracene

The effects of these specific PAHs on human health are discussed below.

Benzo(a)anthracene

Benzo(a)anthracene is classified as a B2 probable human carcinogen based on sufficient data from animal bioassays and no human data. Benzo(a)anthracene produced tumors in mice exposed orally, dermally, and by injection. It also produced mutations in bacteria and in mammalian cells, and transformed mammalian cells in culture (Hazardous Substances Data Bank [HSDB] 2008).

Benzo(b)fluoranthene

Benzo(b)fluoranthene is classified as a B2 probable human carcinogen based on sufficient data from animal bioassays and no human data. It produced tumors in mice after lung implantation, intra-peritoneal and subcutaneous injection, and skin painting. Animals exposed to benzo(b)fluoranthene via dermal and inhalation exposure have been observed to develop skin and lung cancer (HSDB 2008).

Benzo(a)pyrene

Benzo(a)pyrene is listed as a B2 probable human carcinogen based on sufficient animal toxicity data and no human toxicity data. Benzo(a)pyrene has been shown to cause cancer in all of the numerous animal species tested to-date. It has been shown to cause tumors at the site of administration, as well as systemically, via oral, inhalation, and dermal exposure. Exposure to benzo(a)pyrene by ingestion or inhalation may cause mucous membrane irritation, dermatitis, bronchitis, cough, dyspnea, conjunctivitis, photosensitization, pulmonary edema, reproductive effects, and leukemia (HSDB 2008).

Dibenzo(a,h)anthracene

Dibenzo(a,h)anthracene is classified a B2 probable human carcinogen based on sufficient data from animal bioassays and no human data. It has been shown to produce tumors in mice following oral or dermal exposure, and injection site tumors in several animal species following subcutaneous or intramuscular injection. Dibenzo(a,h)anthracene has also been shown to cause DNA damage and gene mutations in bacteria and several types of mammalian cell cultures (HSDB 2008).

4.0 EXPOSURE ROUTES AND PREVENTION

4.1 EXPOSURE ROUTES

Potential exposure routes for subsurface workers that may contact asbestos in soil at IA-3 is inhalation and incidental ingestion. Potential exposure routes for subsurface workers that may contact PAH in soil at IA-3 include incidental ingestion, inhalation of airborne particulates, and dermal contact. Incidental ingestion could occur from casual hand-to-mouth contact, transfer of asbestos fibers or soil from hands to food, ingestion of airborne asbestos or soil particles that enter the mouth or nasal cavity, or other non-intentional ingestion. Inhalation of COCs could occur if asbestos or soil particles become suspended in the air and are inhaled by the subsurface worker. The PAH in IA-3 are not volatile compounds and, therefore, inhalation of volatiles is not a concern for this RMP. Dermal contact with PAH could occur as a result of touching, handling, or other direct contact between the skin and soil, settling of airborne soil particles on exposed skin surfaces, or other dermal contact.

4.2 EXPOSURE PREVENTION

The Property owner and any subsurface contractors working at the Property are expected to take reasonable and appropriate precautions to ensure that workers, visitors, and the general public are not exposed to subsurface soil in IA-3. All subsurface activities must be consistent with applicable U.S. Occupational Safety and Health Administration (OSHA) laws and regulations.

While the potential exposure routes discussed above could occur at the Property, there are easy-to-perform, common-sense precautions that can be implemented to control exposure, such as:

1. Properly barricade and mark the work area to restrict access.
2. Wear clothes that limit the skin area available for contact, such as gloves, hats, long-sleeved shirts, and long pants.
3. Do not smoke, chew gum, or eat in the work area.
4. Do not apply lotion, sunscreen, ointments, or other substances while in the work area, and wash both the application area and the hands prior to application.
5. Cover cuts, scrapes, and other open skin areas.

6. Wash hands frequently, especially before eating, chewing gum or tobacco, smoking, or other activities that involve contact between the hands and items to be placed in the mouth.
7. Wash hands and other exposed areas, especially those areas with visible dirt, before leaving the work site for extended periods of time.
8. Change work clothes shortly after leaving the Property, especially those work clothes that either have visible dirt or have been made damp with sweat or other liquids. Wash such clothes separately from other clothing prior to wearing again. Gloves and other such items that come into direct contact with the contaminated soil should also be washed.
9. Wash hair and body shortly after leaving the work site for the day.
10. Avoid direct contact between the skin and the contaminated soil.
11. Minimize the time that contaminated soil is exposed. The covering of the soil with plastic, straw, or other material will both greatly reduce the potential suspension of the contaminated soil as dust, the ability of persons to directly contact (such as through touch) the contaminated soil, and rainwater runoff of exposed soils.
12. Implement dust suppression procedures at all times during construction. This includes before and during earthwork activities.
13. Air monitoring is discussed in Section 3.3.

Precautions must also be taken to prevent runoff of soil during intrusive earthwork activities. Silt fences, straw, or other suitable devices must be used to prevent co-mingling of potentially impacted soil with the other soils on the Property.

4.3 AIR MONITORING

Air monitoring is required during the first week of subsurface activities to determine if continued use of respirators is warranted, or to verify that dust suppression will prevent worker exposure to elevated levels of COCs. All air monitoring will be conducted under the guidance of a Certified Industrial Hygienist.

For asbestos, OSHA requires that the employer ensure no employee is exposed to airborne concentrations of asbestos in excess of 0.1 fibers per cubic centimeter of air as an 8-hour time-weighted average. The federal standard for asbestos work set forth in 29 CFR 1910.1001 will be

followed for asbestos air monitoring, respirator use, medical surveillance, and other requirements of the standard.

Air monitoring for PAH will be conducted using OSHA Method 58 (OSHA 1986). The OSHA permissible exposure limits (PEL) for the PAHs are provided in Table 1 (NIOSH 2005).

Table 1
Air Monitoring PELs and Methods

COC	Permissible Exposure Limit
Benzo(a)anthracene	Not available
Benzo(b)fluoranthene	Not available
Benzo(a)pyrene	0.2 mg/m ³
Dibenz(a,h)anthracene	Not available

mg/m³ = milligrams per cubic meter

If concentrations are detected above the permissible exposure limit (PEL), or at any concentration where there is no PEL, all workers within or adjacent to the boundaries of IA-3 will use either:

1. A self-contained breathing apparatus that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode will be used by all employees in or near the boundary of the work area, or
2. A supplied-air respirator that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode in combination with an auxiliary self-contained positive-pressure breathing apparatus.

Use of respirators requires adherence to OSHA Standard 29 CFR Part 1910 Subpart I. Prior to the use of respirators, employees must be medically cleared to use respirators and trained in the proper use, limitations, maintenance, and storage of the devices, in addition to receiving a qualitative or quantitative respirator fit test.

Respirators will continue to be used until further monitoring demonstrates COC concentrations have been reduced to below the PEL, or to non-detectable levels when a PEL is not available.

4.4 EXPOSURE RESPONSE

Exposure to asbestos can cause clinical symptoms such as hoarseness, pain, or soreness of the throat. Should any subsurface worker in IA-3 exhibit these symptoms, they will be referred to an ear, nose, and throat specialist for an examination of the upper respiratory tract.

PAH exposure can result in the following symptoms (HSDB 2008):

- Eyes - Photosensitivity and irritation
- Lungs - Irritation with cough and bronchitis
- Mouth – Leukoplakia (white plaques on mucous membranes and tongue)
- Skin - precancerous lesions enhanced by UV light exposure, cysts, nodules, dermal burns, photosensitivity

In the event that exposure to the COCs has occurred or is believed to have occurred, immediate first aid precautions will include the following:

- Contact with eyes – flush with plenty of water and obtain medical attention.
- Contact with skin – wash with plenty of soap and water and obtain medical attention.
- Inhalation – remove to fresh air and obtain medical attention.
- Ingestion – obtain immediate medical attention.

However, the onset of symptoms should not be the only determining factor in seeking medical attention. Other factors, such as the type and duration of incidental exposure, must be considered. For example, if it is found that a construction worker has repeatedly not followed the prescribed precautions to limit exposure, medical attention is warranted.

5.0 HANDLING OF ENVIRONMENTAL MEDIA

The Property assessments completed have indicated the presence of asbestos, metals, and PAHs in soils at concentrations that potentially are subject to regulations regarding proper disposal. These soils have been covered with several feet of clean fill. However, the following procedures should be followed for any excavation within the boundaries of IA-3. In addition, OEPA will be notified of any on- or off-Property disposal of soils from IA-3.

Any soils excavated from the upper two feet may be reused as fill, provided they are not co-mingled with soils from below that depth or impacted by releases of petroleum or hazardous substances subsequent to the date of this plan. The original grade must be retained or raised by replacing the excavated soils with the materials excavated from that depth interval, or with other clean fill. If soils are to be re-used, they must be isolated and temporarily stockpiled separately from other soils.

Soils excavated from below a depth of two feet may contain environmental contaminants. As such, they may be re-used as engineered fill the same area from which they were excavated or properly characterized and disposed of in accordance with applicable laws and regulations.

Table 2 presents the metals analyzed for IA-3 soil, the maximum detected concentration of each, the characteristic hazardous waste thresholds, and the thresholds multiplied by a factor of 20. Based on the analytical data, there is a potential for the soil to be a characteristic hazardous waste for chromium and lead. As a rule of thumb when evaluating likely disposal options, it is assumed that 5% of the soil contaminant will be leachable from the soil therefore, the hazardous waste threshold is multiplied by 20 and the detected concentrations compared to that number.

Table 2
Metals Characteristic Hazardous Waste Data and Thresholds

Chemical	Maximum Detected Concentration (mg/kg)	Hazardous Waste Threshold (mg/L)	20 x Hazardous Waste Threshold (mg/L)
Arsenic	8.07	5	100
Barium	457	100	2,000
Cadmium	Not detected at 5.9	1	20
Chromium	271¹	5	100
Lead	1,150¹	5	100
Mercury	0.71	0.2	4
Selenium	0.42	1	20
Silver	Not detected at 7.8	5	100

mg/kg = milligram per kilogram mg/L = milligram per liter

¹ Exceeds Rule-of-Thumb Threshold Value

6.0 RMP IMPLEMENTATION

This RMP will be implemented when performing any work in the area defined as IA-3. Specifically, this RMP must be implemented and enforced whenever excavation work will be performed in IA-3 to a depth greater than two feet. This RMP is not required to be implemented

when conducting soil excavation on any other portion of the Property, as the applicable standards have been met beyond the boundaries of IA-3.

This RMP must be provided to all contractors, their subcontractors, and their employees who may be involved with subsurface work in IA-3. In addition, this RMP must be provided to any other workers or site visitors present at or immediately adjacent to IA-3 during subsurface work. Contractors, their subcontractors, and their employees must be notified prior to commencement of subsurface work to allow sufficient time to implement the precautions required by the RMP.

7.0 ANNUAL NOTIFICATION

In the event this RMP is implemented, the Property owner will notify the Director of the OEPA by the end of the calendar year in which the RMP was implemented. This notification will be made on an annual basis and will include a discussion of the following:

- The events that required implementation of the RMP.
- The exposures to COCs that may have occurred, including air monitoring results.
- Any on- or off-Property management of soils from IA-3.
- The risk mitigation measures undertaken in accordance with the RMP.

8.0 CRITERIA FOR TERMINATION OF RMP

This RMP can be terminated, with written approval from the Director of the OEPA, upon demonstration that subsurface soil deeper than two feet below grade in IA-3 meet the applicable standards.

9.0 REFERENCES

Hazardous Substances Data Bank (HSDB). 2008. National Library of Medicine on-line database. Website address: <http://toxnet.nlm.nih.gov>.

National Institute of Occupational Safety and Health (NIOSH). 2005. "NIOSH Pocket Guide to Chemical Hazards – Coal Tar Pitch Volatiles." September.

Ohio Environmental Protection Agency (OEPA). 2002. "Voluntary Action Program." Ohio Revised Code Chapters 3745 and 3746.

U.S. Geological Survey (USGS). 1988. "Cincinnati East 7.5-Minute Topographic Map." July.

U.S. Occupational Safety and Health Administration (OSHA). 1986. "Method 58: Coal Tar Pitch Volatiles (CTVP), Coke Oven Emissions (COE), Selected Polynuclear Aromatic Hydrocarbons (PAHs)." July.



DELOME

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Scale 1 : 25,000

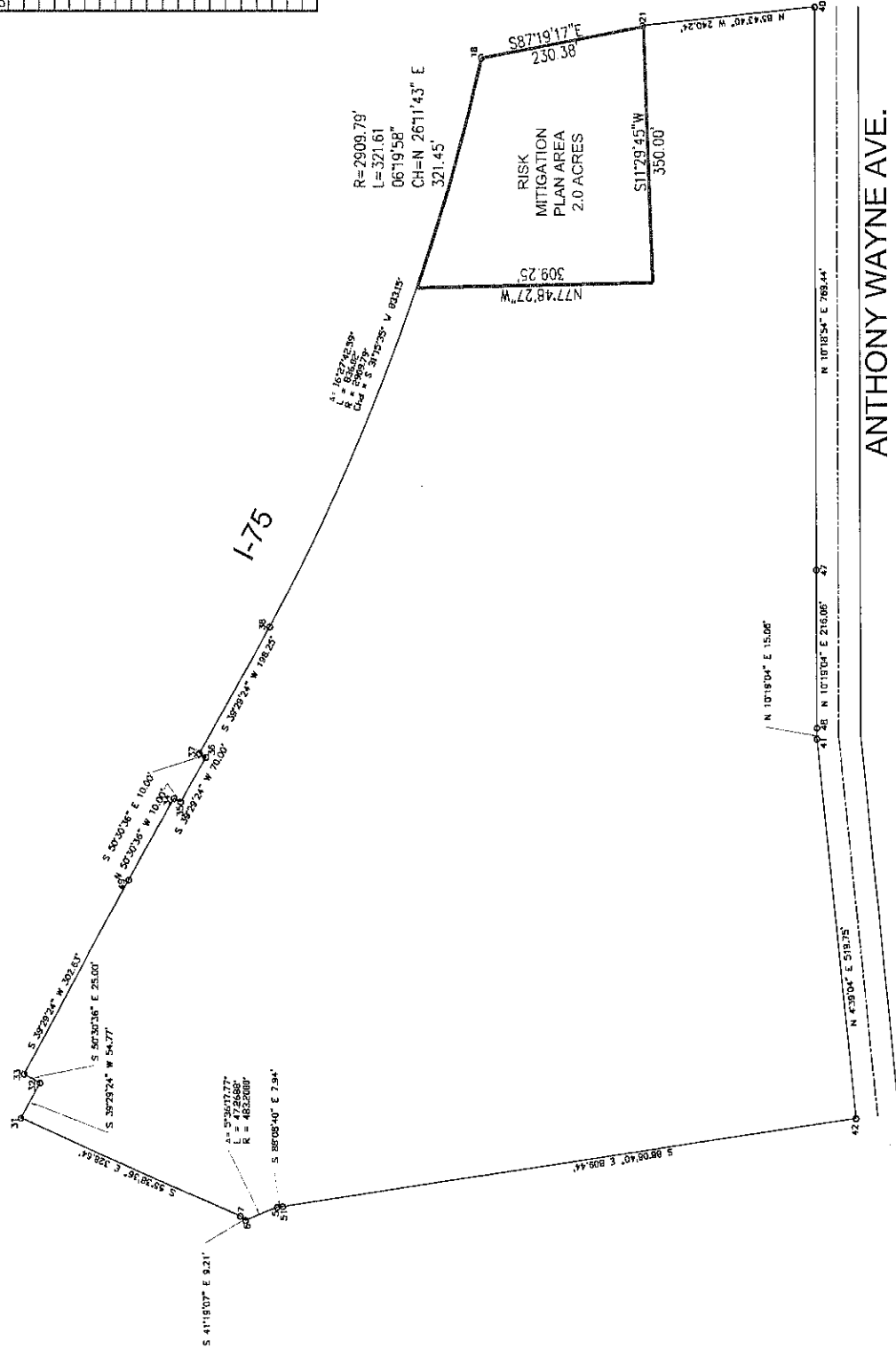
1" = 2080 ft



Figure 1
Location Map
320 South Wayne Avenue
Lockland, Ohio

Prepared by: PDC
Approved by: IM
Date: 7/15/2009
Project Number 2-11708

DATA POINT	LATITUDE	LONGITUDE
5	N 39° 13' 19.15877"	W 84° 27' 29.86568"
6	N 39° 13' 19.26285"	W 84° 27' 29.28039"
7	N 39° 13' 19.19591"	W 84° 27' 29.20123"
8	N 39° 13' 04.29237"	W 84° 27' 36.67348"
9	N 39° 13' 04.34935"	W 84° 27' 39.59000"
10	N 39° 13' 17.42128"	W 84° 27' 25.70537"
11	N 39° 13' 18.99613"	W 84° 27' 26.13423"
12	N 39° 13' 16.04318"	W 84° 27' 25.88475"
13	N 39° 13' 13.50528"	W 84° 27' 29.26737"
14	N 39° 13' 13.56647"	W 84° 27' 29.38718"
15	N 39° 13' 13.02209"	W 84° 27' 29.81782"
16	N 39° 13' 12.98190"	W 84° 27' 29.81602"
17	N 39° 13' 11.42297"	W 84° 27' 31.37753"
18	N 39° 13' 04.35270"	W 84° 27' 42.64889"
19	N 39° 13' 14.11681"	W 84° 27' 40.64438"
20	N 39° 13' 19.24508"	W 84° 27' 40.26192"
21	N 39° 13' 11.66107"	W 84° 27' 41.10750"
22	N 39° 13' 13.96884"	W 84° 27' 40.67456"
23	N 39° 13' 14.49395"	W 84° 27' 28.26545"
24	N 39° 13' 19.15959"	W 84° 27' 29.98654"



ANTHONY WAYNE AVE.

PROJECT NO.	DESCRIPTION	DRAWN BY	CHECKED BY	DATE	ENGINEER SEAL	CONFIDENTIAL - ALL RIGHTS RESERVED - PROPERTY OF PETRO SURVEYING INC., LLC 2000 W. 10th St., Suite 200, Tulsa, OK 74103 Phone: 918.436.4524 Fax: 918.436.4525 www.petro-surveying.com	PREPARED FOR	LOCKLAND	PROPERTY SURVEY	LOCKLAND - 12" x 30"
										REVISION
										INTEGRITY DRAWING
										SHEET NO. 1

Revised Section G of the NFA Checklist

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

Required Information or Component of NFA Letter	Provide Response Below.
1. OAC 3745-300-07 and OAC 3745-300-10 General Information and Description of Geology/Hydrogeology	
1.a. Is the property the subject of an Urban Setting Designation (USD) approved under OAC 3745-300-10(D)? If YES, provide reference where verification of the USD has been demonstrated in accordance with OAC 3745-300-10 (D)(3).	☐ Yes Date USD approved by Director: ☒ No Document: Section: Page Number:
1.b. Are Class V wells a source of contamination? See OAC 3745-300-15(H).	☐ Yes If YES, where is this documented: ☒ No Document: Section: Page Number
1.c. Provide the location of the evaluation of regional geologic/ hydrogeologic and physical characteristics in accordance with OAC 3745-300-07(D)(1)(b).	Document: BHE Phase II Property Assessment Section: 5.3 Page Number: 11 Document: Petro Phase II Property Assessment Adeendum Section: 2.2 Page Number: 2
1.d. Provide the location of the evaluation of property-specific geologic/hydrogeologic characteristics in accordance with OAC 3745-300-07(D)(1)(f).	Document: BHE Phase II Environmental Site Assessment Section: 5.3.2 Page Number: 12 Document: Petro Phase II Property Assessment Adeendum Section: 7.0 Page Number: 10
1.e. Provide the location of the boring and well logs and well construction details in the NFA documentation.	Document: BHE Phase II Property Assessment Section: Figures, Appendix C Page Number: Figures 2 & 3, Appendix C
1.f. Provide the location of the cross-section (a minimum of one) depicting the geology/hydrogeology beneath the property. This cross-section should contain information based on both property-specific as well as regional information that includes all hydrogeologic units important for the Phase II investigation. This includes underlying saturated zones that require protection in accordance with OAC 3745-300-10(E). (See Section D, Question 2.a)	Location of cross-section: Document: BHE Phase II Property Assessment Section: Figures Page Number: Figure 4
1.g. Provide a summary of ground water sampling results in a table. (See Section D, Table 7 for an example.)	Document: BHE Phase II Property Assessment Section: Tables Page Number: Tables 2 & 3
1.h. Provide the location of the evaluation of both on and off-property ground water use for potable purposes in accordance with OAC 3745-300-07(D)(8) and OAC 3745-300-10(C)(2)(a).	Document: Petro Phase II Property Assessment Addendum Section: Appendix 7 Page Number: Entire Appendix

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

2. OAC 3745-300-01(A); OAC 3745-300-07(D)(7) Determining Whether the Water Saturated Zones Is Ground Water

Complete the following for all saturated zones (as appropriate) identified beneath the property.

Identify the Saturated Zone	Depth (top and bottom)	Is the water in the zone ground water per the definition in OAC 3745-300-01(A)(19)? (Provide location where this is documented.)
Zone: Lower Saturated Zone	Top: 100 feet Bottom: 180 feet	☒ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone: Upper Saturated Zone	Top: 19 feet Bottom: 40 feet	☒ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone:	Top: Bottom:	☐ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone:	Top: Bottom:	☐ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone:	Top: Bottom:	☐ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone:	Top: Bottom:	☐ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:
Zone:	Top: Bottom:	☐ Yes If YES , complete question 3 for this zone. ☐ No If NO , check all that apply: ☐ Yield less than 1.5 gallons in 8 hours ☐ Hydraulic conductivity less than 5×10^{-6} cm/sec Document: Section: Page Number:

Continue with Section G for any zone that contains ground water.

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

3. OAC 3745-300-07 (D)(3) Determining Whether Ground Water Meets or Exceeds Unrestricted Potable Use Standards (UPUS)

Complete the following for each zone that meets the definition of ground water.

Ground Water Zone	Complies with UPUS or Exceeds UPUS**	How was the determination of compliance with UPUS made?	What standards were used to determine compliance with UPUS? (Check all that apply)	Reference
Zone: Lower Saturated Zone	☒ Complies with UPUS ☐ Exceeds UPUS	☐ Ground water sampling in accordance with OAC 3745-300-07(D)(3)(a) ☒ Weight-of-evidence demonstration in accordance with OAC 3745-300-07(D)(3)(b) (See Note.)	☒ Generic standards in accordance with OAC 3745-300-08(C) ☐ Standards derived through a Human Health Property-Specific Risk Assessment in accordance with OAC 3745-300-09(D) ☐ Background Levels in accordance with OAC 3745-300-07(H)	Document: BHE Phase II Property Assessment Section: 5.6 Page Number: 14 – 15 Document: Petro Phase II Property Assessment Addendum Section: 8.0 Page Number: 11 – 15
Zone: Upper Saturated Zone	☐ Complies with UPUS ☒ Exceeds UPUS	☒ Ground water sampling in accordance with OAC 3745-300-07(D)(3)(a) ☐ Weight-of-evidence demonstration in accordance with OAC 3745-300-07(D)(3)(b) (See Note.)	☒ Generic standards in accordance with OAC 3745-300-08(C) ☐ Standards derived through a Human Health Property-Specific Risk Assessment in accordance with OAC 3745-300-09(D) ☐ Background Levels in accordance with OAC 3745-300-07(H)	Document: BHE Phase II Property Assessment Section: 5.4 Page Number: 12
Zone:	☐ Complies with UPUS ☐ Exceeds UPUS	☐ Ground water sampling in accordance with OAC 3745-300-07(D)(3)(a) ☐ Weight-of-evidence demonstration in accordance with OAC 3745-300-07(D)(3)(b) (See Note.)	☐ Generic standards in accordance with OAC 3745-300-08(C) ☐ Standards derived through a Human Health Property-Specific Risk Assessment in accordance with OAC 3745-300-09(D) ☐ Background Levels in accordance with OAC 3745-300-07(H)	Document: Section: Page Number:
Zone:	☐ Complies with UPUS ☐ Exceeds UPUS	☐ Ground water sampling in accordance with OAC 3745-300-07(D)(3)(a) ☐ Weight-of-evidence demonstration in accordance with OAC 3745-300-07(D)(3)(b) (See Note.)	☐ Generic standards in accordance with OAC 3745-300-08(C) ☐ Standards derived through a Human Health Property-Specific Risk Assessment in accordance with OAC 3745-300-09(D) ☐ Background Levels in accordance with OAC 3745-300-07(H)	Document: Section: Page Number:
Zone:	☐ Complies with UPUS ☐ Exceeds UPUS	☐ Ground water sampling in accordance with OAC 3745-300-07(D)(3)(a) ☐ Weight-of-evidence demonstration in accordance with OAC 3745-300-07(D)(3)(b) (See Note.)	☐ Generic standards in accordance with OAC 3745-300-08(C) ☐ Standards derived through a Human Health Property-Specific Risk Assessment in accordance with OAC 3745-300-09(D) ☐ Background Levels in accordance with OAC 3745-300-07(H)	Document: Section: Page Number:

Note: A weight-of-evidence demonstration cannot be used to determine that ground water exceeds UPUS.

****Complete question 4 for zones that comply with UPUS. Complete questions 5 through 8 for zones that exceed UPUS.**

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

Required Information or Component of NFA Letter	Provide Response Below.	
4. <u>OAC 3745-300-07(D)(4); OAC 3745-300-10(E)</u> Demonstrating Protection of Ground Water Zones Meeting UPUS Questions 4.a through 4.e must be completed for the upper most ground water zone that meets UPUS. Ground water zones beneath this zone may be considered protected from sources on the property unless there is a reason to believe a release of hazardous substances or petroleum has been released to underlying zones.		
4.a. Identify the ground water zone and depth for which this section applies.	Ground Water Zone: Lower Saturated Zone Depth (top): 100	
4.b. Provide the location of the evaluation conducted to ensure that the migration of hazardous substances or petroleum from sources or source areas on the property will not result in UPUS being exceeded anywhere within the ground water zone as required by OAC 3745-300-10(E) and OAC 3745-300-07(D)(4) and specify which demonstration was used.	Specify which demonstration was used (check all that apply): ☐ Generic Leach-based soil values developed by Ohio EPA ☐ Derived Property-specific leaching values ☒ Weight-of-evidence (OAC 3745-300-07(D)(4)(a)(ii)) Document: BHE Phase II Property Assessment Section: 5.6 Page Number: 14 – 15 Document: Petro Phase II Property Assessment Addendum Section: 8.0 and Tables Page Number: 11-15 and Tables IV and V	
4.c. Was a generic dilution factor used as provided in "Ohio EPA Derived Leach-Based Soil Values Technical Guidance Document" (February, 2002), Section G - Attachment I, or was a property-specific dilution factor determined?	☐ Yes If yes, specify which: ☒ No ☐ Generic Dilution Value ☐ Calculated Dilution Factor Dilution Factor:	
4.d. Was a model used to demonstrate protection of ground water meeting unrestricted potable use standards? If yes, provide the name of the model(s) and complete question 9 (Use of Models).	☐ Yes Model Name(s): ☒ No	
4.e. Was/is the remedy implemented necessary in order to prevent leaching of COCs from the property to ground water?	☐ Yes ☒ No	
5. <u>OAC 3745-300-07(D)(8) and OAC 3745-300-10(A)(C)</u> Classification of Ground Water Zones.		
Indicate Each Saturated Zone That Exceeds UPUS	Indicate the Classification and Yield of the Zone	Indicate the Basis For Classification of the Ground Water Zone (Provide the location of the documentation for Classification of the Zone)
Zone: Upper Saturated Zone	☐ Critical Resource ☐ Class A ☒ Class B YIELD: 2.19 GPM	Check all that apply: ☐ Within a Drinking Water Source Protection Area for a Public Water System Using Ground Water ☐ Within a Consolidated Sole Source Aquifer ☐ Presence of Potable Wells Within ½ Mile of the property boundary ☐ Total Dissolved Solids ☐ Yield determined by ODNR ground water resource maps or other published and verified data in accordance with OAC 3745-300-07 (D)(8)(b)(i). (Note: This option cannot be used to determine the zone is Class B ground water.) ☒ <i>In-situ</i> property-specific yield determination conducted in accordance with OAC 3745-300-07(D)(8)(b)(ii) or (D)(8)(c) and OAC 3745-300-07(D)(9). Document: Petro Phase II Property Assessment Addendum Section: 7.0 Page number: 10-11

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

Zone: Lower Saturated Zone	☐ Critical Resource ☐ Class A ☐ Class B YIELD: 800-1400 GPM	Check all that apply: ☐ Within a Drinking Water Source Protection Area for a Public Water System Using Ground Water ☐ Within a Consolidated Sole Source Aquifer ☐ Presence of Potable Wells Within ½ Mile of the property boundary ☐ Total Dissolved Solids ☐ Yield determined by ODNR ground water resource maps or other published and verified data in accordance with OAC 3745-300-07 (D)(8)(b)(i). (Note: This option cannot be used to determine the zone is Class B ground water.) ☐ <i>In-situ</i> property-specific yield determination conducted in accordance with OAC 3745-300-07(D)(8)(b)(ii) or (D)(8)(c) and OAC 3745-300-07(D)(9). ☒ Not Applicable. Because of demonstration that the lower zone meets UPUS Document: Section: Page number:
Zone:	☐ Critical Resource ☐ Class A ☐ Class B YIELD:	Check all that apply: ☐ Within a Drinking Water Source Protection Area for a Public Water System Using Ground Water ☐ Within a Consolidated Sole Source Aquifer ☐ Presence of Potable Wells Within ½ Mile of the property boundary ☐ Total Dissolved Solids ☐ Yield determined by ODNR ground water resource maps or other published and verified data in accordance with OAC 3745-300-07 (D)(8)(b)(i). (Note: This option cannot be used to determine the zone is Class B ground water.) ☐ <i>In-situ</i> property-specific yield determination conducted in accordance with OAC 3745-300-07(D)(8)(b)(ii) or (D)(8)(c) and OAC 3745-300-07(D)(9). Document: Section: Page number:
Note: Critical Resource ground water is defined in OAC 3745-300-10(C)(1). Critical Resource Ground Water - The ground water is being used by a public water system and is in a drinking water source protection area for a public water system using ground water as provided in OAC 3745-300-10 (C)(1)(a). - The ground water is part of an unconsolidated saturated zone capable of yielding greater than 100 gpm over 24 hours as provided in OAC 3745-300-10 (C)(1)(b). - The ground water is part of a consolidated saturated zone that is part of a sole source aquifer as provided in OAC 3745-300-10(C)(1)(c). - It has been assumed to be critical resource for purposes complying with ground water response requirements as provided in OAC 3745-300-10(B)(2).		
Note: Class A ground water is described in OAC 3745-300-10(C)(2). A ground water zone can only be classified as Class A if it doesn't meet the criteria for Critical Resource ground water. Class A Ground Water - The ground water is in a saturated zone currently utilized for potable purposes on or within ½ miles of the property boundary as provided in OAC 3745-300-10(C)(2)(a). - The background level of total dissolved solids is less than 3,000 mg/L. - The ground water is in a saturated zone that is capable of yielding greater than 0.1 gpm over 24 hours as provided in OAC 3745-300-10(C)(2)(b). - The ground water does not meet any of the criteria making it critical resource and ground water response requirements for Class A ground water have been followed.		
Note: Class B ground water is described in OAC 3745-300-10(C)(3). A ground water zone can only be classified as Class B if it doesn't meet the criteria for Critical Resource or Class A ground water. Class B Ground Water - The background level of total dissolved solids is equal to or greater than 3,000 mg/L. - The zone yields less than 0.1 gpm over 24 hours. - The zone yields less than 3 gpm over 24 hours and another ground water zone meeting the requirements listed in OAC 3745-300-10(C)(2)(c)(i) underlies the property. - The zone yields less than 3 gpm over 24 hours and all parts of the zone are wholly contained within 15 feet below ground surface.		

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

Required Information or Component of NFA Letter	Provide Response Below.
6. OAC 3745-300-07(D)(6)(d) Determination of the Concentration of Chemicals of Concern in Ground Water	
6.a. Reference the location of the documentation describing how the criteria required for sampling of ground water in accordance with OAC 3745-300-07 (D)(6)(d)(i) through (D)(6)(d)(vi) have been met. Provide a summary of ground water sampling results in a table. (See Section D, Table 7 for an example.)	Document: BHE Phase II Property Assessment Section: 4.2.2 Page Number: 8 - 9 Location of table: Tables 2 & 3
6.b. Provide the location where all data collection, activities, sampling procedures and rationale for the sampling and testing activities are summarized in accordance with OAC 3745-300-07(D)(6)(d) and 07(E)(1) and (3).	Document: BHE Phase II Property Assessment Section: 4.2.2 Page Number: 9 (last paragraph)
7. OAC 3745-300-07(D)(10) Determination of Off-Property Sources or Source Areas Impacting Ground Water	
7.a. Are there sources or source areas located off the Property that contributed to or are contributing to ground water contamination on the Property? If NO , proceed to 8.	☐ Yes ☒ No
7.b. If YES to 7.a, is the volunteer or owner/operator associated with any off-property source or source areas contributing to the ground water contamination on the Property as provided in OAC 3745-300-10(F)(1)(b)(i) through (F)(1)(b)(v)? If YES , proceed to 8.	☐ Yes ☐ No
Note: If YES , applicable ground water response requirements are the same as if all sources are on the property as specified in OAC 3745-300-10(F)(1)(c).	
7.c. If NO to 7.b, are the chemicals of concern (COCs) from the off-property sources or source areas the same as or different from the COCs from on-property sources or source areas? Provide a reference where this evaluation is discussed. The discussion should indicate which COCs are from off-Property sources and whether they are the same or different from the COCs on the voluntary action property and how this determination was made. This needs to be evaluated separately for each Identified Area and must indicate how the requirements in OAC 3745-300-07(D)(10) have been met.	Select One: ☐ COCs all from off-Property sources. ☐ COCs from off-Property sources are same as COCs from on-Property sources. ☐ COCs from off-Property sources are different from COCs from on-Property sources. ☐ There is a combination where some COCs from off-Property sources are the same and some are different from the on-Property sources of COCs. Document: Section: Page Number:
8. OAC 3745-300-10 (F) Ground Water Response Requirements	
Based on the presence or absence of an urban setting designation, ground water classification, and whether COCs are from on and/or off-property sources, indicate the appropriate rule citation for the ground water response requirements for each zone. [See attached worksheet and flowchart at the end of this section - e.g., for Critical Resource ground water, No USD, all Sources on the Property, the rule citation is OAC 3745-300-10 (F)(2)(a)] Provide the location of the documentation which shows how the ground water response requirements were met.	Provide the appropriate rule citation for the ground water response requirements for each ground water zone that exceeds UPUS: Zone: Upper Saturated Zone Citation: 3745-300-10 (F)(4)(a) Zone: Lower Saturated Zone Citation: 3745-300-10 (F)(6)(a) Zone: Citation: Document: Petro Phase II Property Assessment Addendum Section: 8.0 Page Number: 11 - 14
Note: If Class V wells are the source of ground water contamination, see OAC 3745-300-15(H).	

SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

9. OAC 3745-300-07(F) Use of Modeling

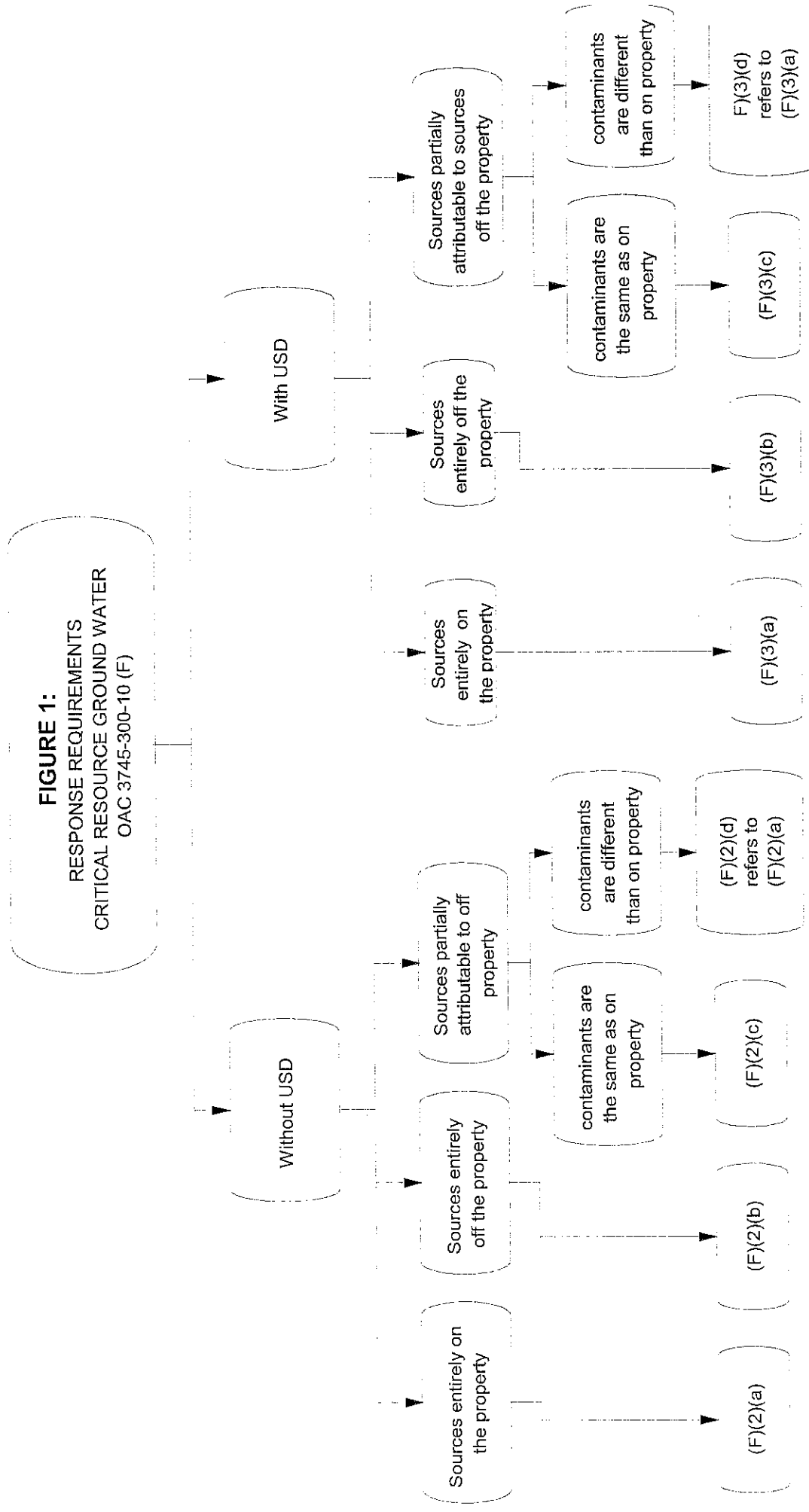
Complete questions 9.a. through 9.d. for each model used to evaluate the ground water or leaching to ground water. Make copies as necessary.

9.a. Was ground water modeling or modeling to evaluate leaching conducted? If YES , indicate what was the purpose of the modeling exercise and then proceed with 9.b below. Otherwise, do not complete this section.	☒ Yes Purpose of Model: Predict whether UPUS were exceeded at point-of-compliance ☐ No Model Name:AT123D Model Version: 6, Dated 2001
9.b. Identify the ground water zone(s) for which modeling was conducted and provide a reference for the modeling report. A modeling report addressing OAC 3745-300-07 (F)(1) through (F)(4) and the modeling results must be provided in accordance with OAC 3745-300-07(F)(5).	Ground Water Zone(s): Upper Saturated Zone Depth(s): Location of Modeling Report: BHE Phase II Property Assessment Appendix I Petro Phase II Property Assessment Addendum Appendix 8

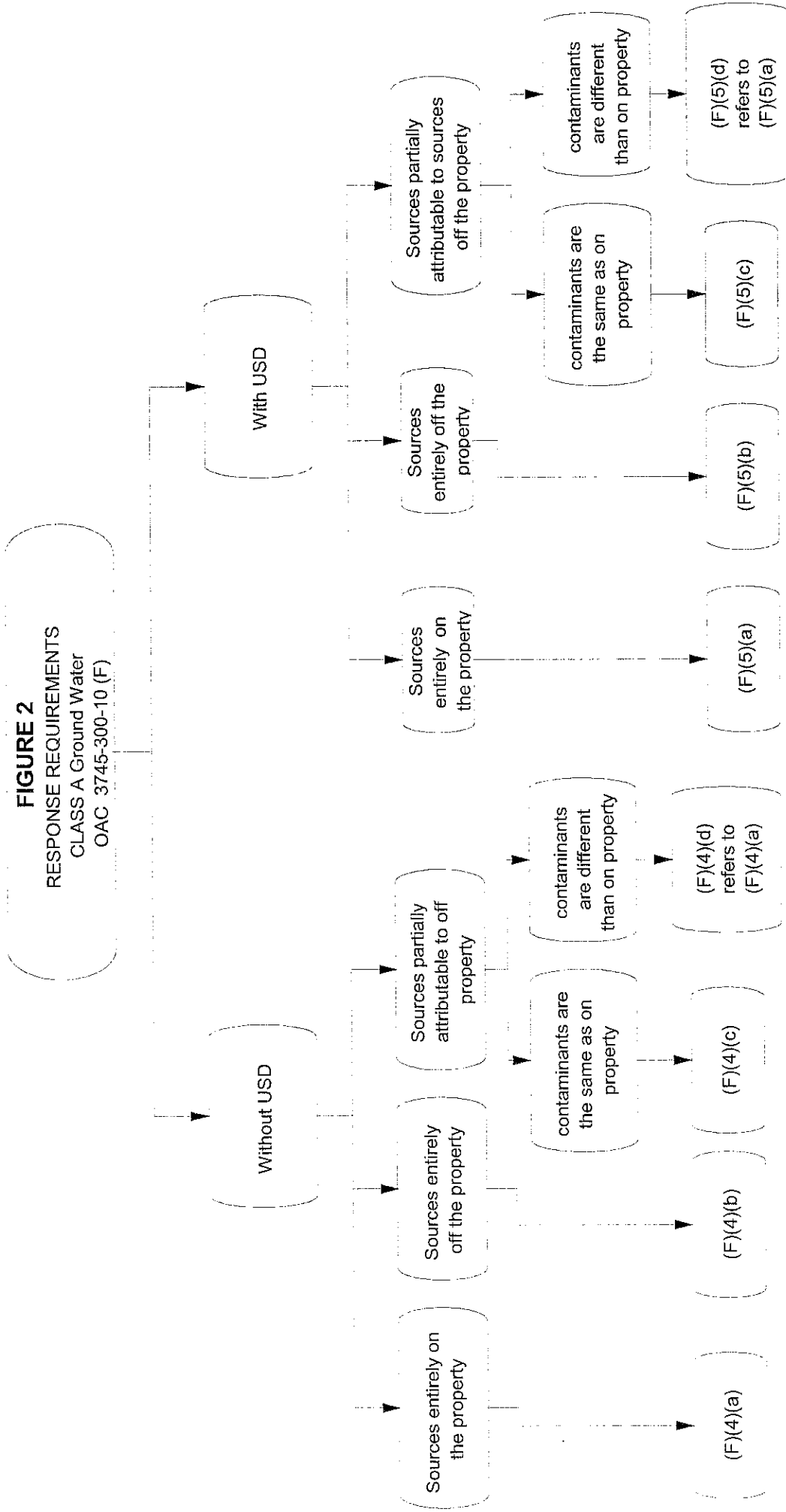
SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER

Note: Pursuant to OAC 3745-300-07(F)(5), the following must be summarized in the Modeling Report: - The purpose of the modeling exercise(s). - The applicability of the model used for the modeling exercise(s) and how the model complies with OAC 3745-300-07(F)(2). - The assumptions used in the modeling exercise(s). - The input parameters used in the modeling exercise(s) and the basis for their use. - If necessary, method(s) used to calibrate and/or field validate the model(s). - The sensitivity of the model(s) to input parameters and how the modeling complies with OAC 3745-300-07(F)(4)(a) and (F)(4)(b). - The results and conclusions drawn from the modeling exercise(s).	
9.c. Attach a table providing a list of modeling parameters, input values, source of the data, input range (if uncertainty analysis conducted). (See Section D, Table 9 for an example.)	Location of Table: BHE Phase II Property Assessment Section: Appendices Page: Appendix I Tables 1 & 2 Location of Table: Petro Phase II Property Assessment Addendum Section: Tables Page: Tables VI
Note: If a degradation rate was used, the modeling report must justify the use of this input parameter.	
9.d. Is the presence of an engineering control or an institutional control used to adjust input parameters? If YES, reference where this is documented.	☐ Yes ☒ No
Note: Pursuant to OAC 3745-300-15, engineering controls must be addressed in O&M plan and institutional controls must be recorded in the same manner as a deed.	

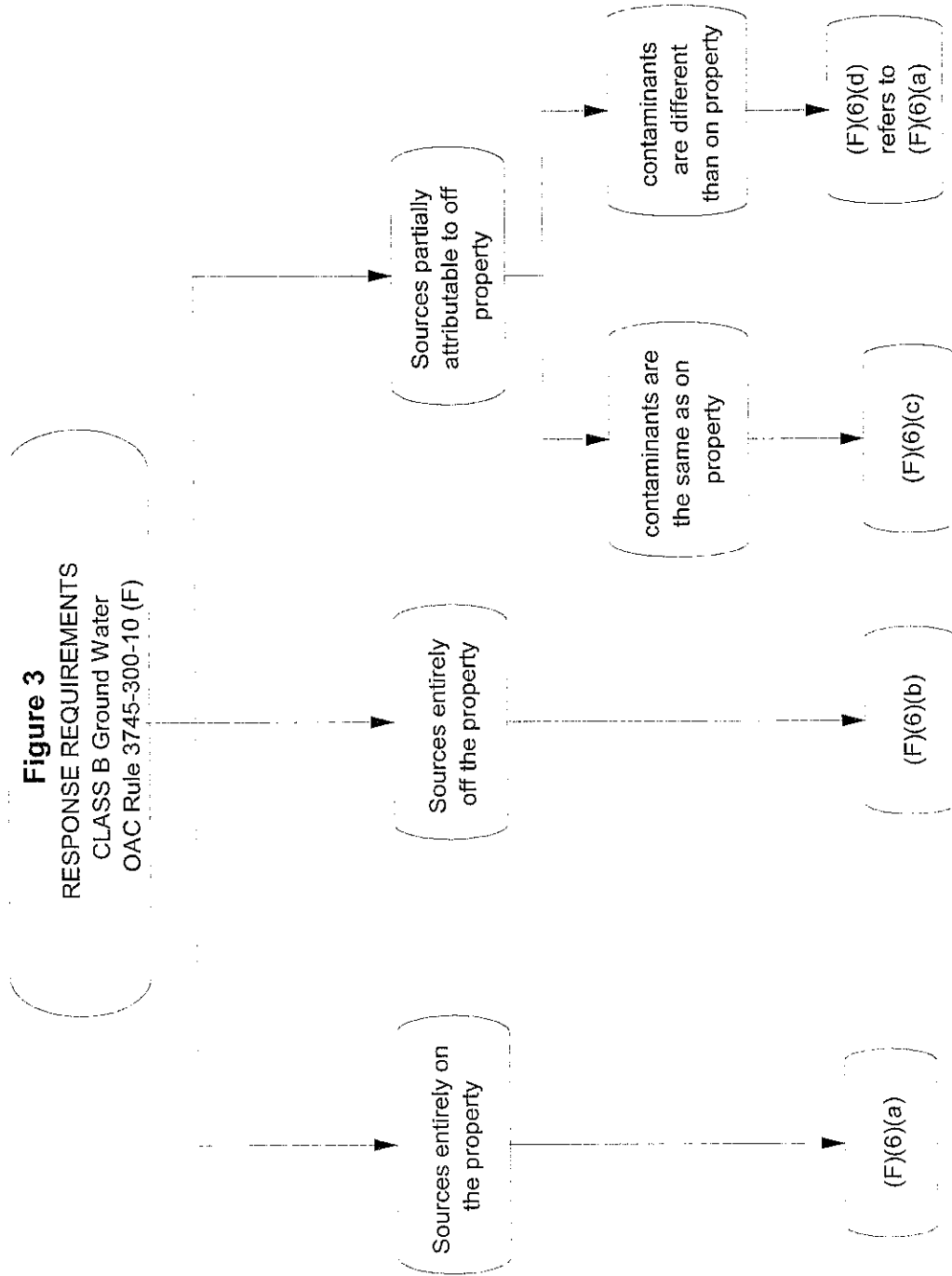
SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER



SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER



SECTION G: PHASE II PROPERTY ASSESSMENT-GROUND WATER



END OF SECTION G

Article: "Field Comparison of Micropurging vs. Traditional Ground Water Sampling"

CONF-950868-14

Evaluation of Micropurging Versus Traditional Groundwater Sampling at the Department of Energy's Kansas City Plant

Kansas City Division

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EVALUATION OF MICROPURGING VERSUS TRADITIONAL GROUNDWATER SAMPLING AT THE DEPARTMENT OF ENERGY'S KANSAS CITY PLANT

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EVALUATION OF MICROPURGING vs. TRADITIONAL GROUNDWATER SAMPLING AT THE DEPARTMENT OF ENERGY'S KANSAS CITY PLANT

by

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ABSTRACT

A field trial comparing the micropurge and the traditional purge and sample method of groundwater sampling was conducted at the U.S. Department of Energy (DOE), Kansas City Plant (KCP) in 1993 and 1994. Duplicate groundwater samples were collected using traditional and micropurge methods, analyzed for selected organic and inorganic constituents, and the results compared statistically. Evaluation of the data using the Wilcoxon Sign Rank test indicates that within a 95% confidence interval, there was no significant difference between the two methods for the site contaminants and the majority of naturally occurring analytes. These analytical results were supported by visual observations with the colloidal borescope, which demonstrated impacts on the flow system in the well when using traditional sampling methods. Under selected circumstances, the results suggest replacing traditional sampling with micropurging based on reliability, cost, and waste minimization.

INTRODUCTION

Traditional sampling practices for groundwater monitoring wells commonly consist of purging three or more well volumes prior to sample collection. The rationale for purging wells is to remove stagnant water from the wellbore in order to permit representative water from the surrounding porous medium to enter the well.

The alternative sampling approach using micropurging is based on the premise that stagnant water in the well casing does not mix with groundwater flowing through the screen. Using tracer experiments, Robin and Gillham (1987) showed that groundwater moves through the screened portion of a well with little interaction or mixing with stagnant water in the

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overlying well casing. Similar studies by Powell and Puls (1993) supported this observation. These studies suggest that flow in the well screen is horizontal and laminar.

Using the colloidal borescope Kearn et al. (1992) suggested that water samples taken directly from the screened interval with a dedicated sample device were representative of groundwater in the surrounding formation and that purging several well volumes prior to sample collection is unnecessary. Mixing of the stagnant water with formation water in the screened interval of the well is avoided when the dedicated sample device is left in place such that flow in the screened interval is not disturbed. By using low pumping rates, only fresh formation water and only particles that are being advected by groundwater flow are sampled. Sampling at low flow rates without filtering, therefore, allows quantification of the total "mobile" pollutant load. In other words, both the contamination that is moving in the dissolved phase and contamination sorbed to mobile particles is being measured.

To determine whether groundwater samples collected by the micropurge methodology had application at the KCP, a site with very low flow rates and slightly reducing conditions, a comparison study was conducted. Consequently, samples were collected at low rates, (~ 100 ml/min); using a dedicated sampling pump to prevent disturbing the well. Only the sample tubing and pump are purged. The purging of three or more wellbore volumes associated with traditional sampling is eliminated, as are the large volumes of purge water requiring disposal. The two sampling methods were statistically compared to determine if differences exist.

Twelve sample events were conducted from February 1993 to March 1994 on six monitor wells. Wells were selected based on historical VOC concentrations to test the micropurge methodology on two predominantly clean wells (<100 ppb total VOCs); two moderately contaminated wells (100-1500 ppb VOCs); and a highly contaminated well (>1500 ppb VOCs). In addition, all selected wells had moderate to high values of inorganic compounds such as iron (5-45 mg/L), calcium (150-200 mg/L), magnesium (5-25 mg/L), and manganese (2-11 mg/L).

METHOD

Dedicated submersible pumps were installed in monitor wells with pump screens installed in the mid-portion of the well screen. Pumps were precalibrated to pump at 100 - 200 ml/min. The pumps were suspended in the middle portion of the screened interval by stainless steel wire rope fitted with 1/4" ID teflon discharge tubing and secured to a stainless steel quick connect mounted in a PVC well cap. The sample line discharge "pigtail" consisted of 1/4" ID Teflon tubing and was affixed to the quick connect in the PVC caps during sample events. The customized well caps were mounted snugly into the two inch PVC well casing preventing any unwanted motion of the pump when sampling.

Micropurge sampling was conducted initially followed by the traditional method of purging three bore volumes prior to sample collection. Data collected in the field (i.e., depth to water, pH, conductivity, temperature, volume of water prepumped, and pertinent well physical information) was recorded. For both, specific sampling methodology is described below.

Twelve test events were conducted over a 13-month period to account for seasonal variations. Previous investigations at the KCP have identified the presence of VOC contaminated groundwater. Metals contamination in site groundwater is not of concern and no inorganic plume development has been identified. Metals were sampled during this study as an additional basis of comparison, however.

Site soils are comprised of clayey silts with groundwater flow rates comparatively low (hydraulic conductivity 1.5 - 2.3 ft/day) which magnify the effect of colloids mobilized by high flow rate purging and sampling. Assuming proper well installation, water in this portion of the well is continuously exchanged with that of the surrounding aquifer. Samples collected at low flow rates with dedicated devices draw the sample from the screened portion of the well which is continuously replenished with water from the surrounding aquifer.

Dissolved phase transport is the primary migration component for VOCs. Therefore, minimal differences were expected in a head-to-head comparison of the two sample methodologies. Results of the two sample methodologies were not expected to be comparable for inorganics because the colloidal transport of inorganics is a major migration component and high concentrations of naturally occurring iron and manganese at the Kansas City Plant.

SAMPLING PROCEDURES - SPECIFIC

Micro-purge - The following steps were performed in the order and manner below:

A gas-powered generator powered the pump and controller box. Water discharge rates were slowly adjusted to achieve the desired 100-150 ml/minute discharge rate. Water was continuously extracted at ~100-150 ml/min during sampling. Two volumes of the sample discharge line were purged (~1 liter), VOC samples were collected, followed by collection of filtered and unfiltered inorganics.

Traditional Bailer Sample - The following steps were performed:

The well was purged at ~1 gal/min using the installed dedicated pump. Purging continued until temperature, conductivity, and pH stabilized, or three well bore volumes were extracted. The dedicated pump was then extracted and after the well had recovered a minimum of 95% of the initial static water elevation, sampling was initiated. Using a stainless steel bailer equipped with a bottom emptying device, VOC samples were collected followed by collection of filtered and unfiltered inorganics utilizing a peristaltic pump.

QA/QC

VOC trip and equipment blanks were collected and analyzed for each sample event. Since the micropurge sample methodology utilizes dedicated equipment, equipment blank contamination is not a concern. Other QA/QC steps specific to the lab included matrix spike/matrix spike duplicate (MS/MSD) analysis, laboratory control samples, method blanks, and surrogates.

DISCUSSION OF RESULTS

FIELD STABILIZATION RESULTS

Field stabilization readings of temperature pH and conductivity were collected and analyzed in a flow-through cell, and compared during the study. During the micropurge test, field sampling measurements were not measured immediately upon initiation of purging. The submersible pump and associated lines were purged, then VOCs and inorganics were collected in series. The flow through cell was then filled and approximately two cell volumes purged and stabilization parameters measured for micropurge. For a given well, approximately 4 liters were usually purged prior to obtaining the first field measurement during the micropurge sample event. Field stabilization readings were recorded after three bore volumes (~45 liters/well) had been purged during the traditional sample event.

Stabilization Parameter Comparisons

Comparison of pH, temperature, and conductivity values were compared from the first micropurge field measurement reading to the last reading collected after traditional purging. Values for pH were within plus or minus ten percent with two exceptions (DOE 1994). Temperature values were likewise comparable with the exception of one sampling event (DOE 1994). Freezing temperatures caused sample equipment to get progressively colder at a given well resulting in a lower value for the last reading. Conductivity values were not as comparable, however. Approximately half of the values did not satisfy the plus or minus ten percent criteria with one well (64L) consistently giving values beyond the ten percent stabilization criteria (DOE 1994). Elevated iron concentrations in this well and associated suspended solids during traditional purging accounted for this discrepancy.

Discounting the conductivity results determined in well 64L and temperature results from the March 1993 sample event, field stabilization tests are comparable between the two sample methodologies. Based on the above, results do not indicate the need to purge large volumes of water from the well prior to sample collection.

DATA ANALYSIS

Before presenting the data comparison and statistical review of results from the two sampling methods, two issues must be emphasized. The first is that the contaminants of interest, and the ones for which monitoring is required by EPA at all but one location at the KCP, are the organic compounds. Considerable data exist to demonstrate that there is no inorganic contaminant plume development at the KCP. The second issue of importance is the site's physical characteristics: site soils are clayey-silts, groundwater flow rate is slow, and subsurface conditions are reducing. Slow flow rates and reducing conditions are obviously the circumstances under which it is most difficult to demonstrate the acceptability of micropurging for inorganic constituents primarily because of the redox-sensitive elements (iron and manganese). Redox sensitive elements are prone to significant fluctuations in concentration

when redox conditions change such as when a sample from a slightly reducing environment is brought to the oxygenated surface environment (Kearl et al. 1994).

VOCs and 23 metals were analyzed with each sampling event, however, only those constituents detected are discussed. Following is a well by well discussion of results for organics and total and filtered inorganics. A statistical review of the data follows the well by well comparison.

Data Discussion

Well KC97-58U

Organics - This well was sampled 8 times during the comparison study from February to August 1993. Traditional sampling detected a cumulative total of 544 ppb of organics from the eight sampling events consisting of chloroethene, methylene chloride, 1,2 dichloroethylene (1,2 DCE), trichloroethylene (TCE) and methyl isobutyl ketone (MIBK). Micropurge sampling detected a total of 431 ppb (total) of the same organics.

Inorganics - The following elements were detected: Aluminum (Al), Arsenic (As), Barium (Ba), Cadmium (Cd), Calcium (Ca), Copper (Cu), Iron (Fe), Lead (Pb), Magnesium (Mg), Manganese (Mn), Potassium (K), Sodium (Na) and Zinc (Zn). Aluminum was effectively removed by filtering in all but two samples. Similar concentrations were noted for all other analytes from this well.

Well KC87-64L

Organics - This well was sampled 8 times between February and August 1993 with contaminants consisting of vinyl chloride, 1,1 dichloroethylene (1,1, DCE), 1,2 DCE and TCE. Total VOC results for the 8 sampling events for micropurge totaled 49,373 ppb with traditional sampling totaling 44,380 ppb.

Inorganics - Al, As, Antimony (An), Ba, Cd, Ca, Fe, Pb, Mg, Mn, K, Se, Na, and Zn were detected in this well. Micropurge samples were slightly but consistently higher than traditional sample results for Ba, Ca, Fe, Mg, Mn, and Na. All other compounds exhibited good correlation between sampling methods. Aluminum was effectively filtered out in both sampling methods in all but one case.

Well KC87-76L

Organics - This low level contaminated well was also sampled 8 times between February and August 1993. Micropurge sampling detected organics totaling 273 ppb with traditional sampling from the eight events totaling 237 ppb. Contaminants consisted of vinyl chloride, 1,2 DCE and TCE.

Inorganics - Al, As, Ba, Cd, Ca, Fe, Mg, Mn, K, Na and Zn were detected in this well. Filtering effectively removed Al in all but one sample. Good agreement existed between all other compounds during sampling.

Well KC91-168L

Organics - This contaminated well was sampled four times between March and April 1994 with contamination consisting primarily of 1,2 DCE with one detection of methyl bromide during the first test event. During four sample events total VOCs for the micropurge method equaled 16,370 ppb with the traditional sampling method totaling 7700.

Inorganics - Al, As, An, Ba, Cd, Ca, Cr, Cu, Fe, Mg, Mn, K, Na, and Zn were detected in this well. Traditional sampling produced higher sodium results on average while manganese results for micropurge were slightly higher on average.

Well KC85-34L

Organics - This contaminated well was sampled four times between March and April 1994. Contaminants included only TCE and 1,2 DCE. Total organics for the four sampling events for the micropurge method totaled 23,570 ppb while traditional sampling totaled 21,480 ppb.

Inorganics - Al, As, An, Ba, Cd, Ca, Fe, Mg, Mn, K, and Zn were detected from this well. An overall good agreement was noted between all compounds sampled.

Well KC85-39U

This low level contaminated well was sampled between March and April 1994 and exhibited contamination only by 1,2 DCE. Micropurge sampling from the four sampling events totaled 666 ppb with the traditional method totaling 680 ppb.

Inorganics - Al, As, An, Ba, Cd, Ca, Fe, Mg, Mn, K, Na, and Zn were detected in this well. An overall good agreement was noted among all elements detected. A few sporadic detections of inorganics were noted but no pattern existed as the scattered detections were noted from traditional sampling for some elements and from micropurge for others.

Statistical Review

Table 1 is a statistical comparison of micropurge versus traditional sampling for organic compounds. Table 2 is a statistical comparison of micropurge non-filtered versus traditional filtered inorganic samples. The logic behind this comparison is that the micropurge methodology does not mobilize colloidal sized particles and the traditionally collected filtered sample removes mobilized colloids when a 0.45 micron filter is used.

The objective of the statistical analysis was to evaluate the equivalence of the micropurge sample methodology to the traditional bailer method. Equivalence of the methods was assumed to be that, from a given well on a given day, the values obtained from the two sampling methods have the same distribution. If the two methods are not equivalent, the difference was assumed to be due to sampling heterogeneity. The data collected was intended to compare the methods on this basis. The samples are paired in that they were collected from the same well on the same date and were analyzed on the same date.

The expected results of this study were to observe no significant change in VOC analysis when comparing the two sampling methods as it was assumed that the traditional sampling

TABLE 1
STATISTICAL COMPARISON OF MICROPURGE vs TRADITIONAL ORGANIC SAMPLES WILCOXON RANK

VINYL CHLORIDE		1,2 DCE		TCE	
ORDERED		ORDERED		ORDERED	
DIFFERENCE	RANK	DIFFERENCE	RANK	DIFFERENCES	
-1	2.5	-1	1	-3.5	
-1	2.5	2	2	-50	
-1	2.5	-3	3	-100	
-1	2.5	-4	4.5	-355	
-2	5.5	-4	4.5	500	
-2	5.5	-5	6	600	
-5	7.5	-7	7.5	700	
-5	7.5	7	7.5	-840	
-6	9	10	9	-1140	
7	10.5	-20	10	-1175	
7	10.5	-50	11	-2475	
-9	12.5	100	13	WSR calc.	
9	12.5	100	13	NUM +'s	
10	14	-100	13	sign prob	
11	15	120	15	WSR (.01)	
-17	16	170	16		
80	17	500	17		
150	18	700	18		
220	19	1100	19		
220	20	1700	20		
-460	21	2100	21		
700	22	3100	22		
WSR calc.	-34	-3500	23.5		
NUM +'s	10	3500	23.5		
SIGN PROB	0.588	3600	25		
WSR (.01)	-77.5	5900	26		
		WSR calc.	91.5		
		NUM -'s	10		
		sign prob	0.1634		
		WSR (.01)	99.5		

TABLE 2
STATISTICAL COMPARISON OF MICROPURGE vs TRADITIONAL FILTERED INORGANIC SAMPLES

E L E M E N T	N	M E A N	S T D	T	P R O B T	M S I G N	P R O B M	S I G N R A N K	P R O B S	N O R M	P R _ N	U C L 9 7 5	L C L 9 7 5
Al	31	0.4431	0.7165	3.44317	0.0017	12.5	0	215	0	0.85402	0.0004	0.7059	0.1803
As	29	0.0388	0.096	2.17757	0.038	8	0.0037	125.5	0.0025	0.65689	0.0	0.0753	0.0023
Ba	36	0.1334	0.2062	3.88024	0.0004	11	0.0003	241.5	0	0.81406	0	0.2031	0.0636
Ca	36	7.8389	16.7067	2.81523	0.008	8	0.0113	158	0.0109	0.93026	0.0318	13.492	2.1861
Fe	36	10.069	14.1612	4.26597	0.0001	15	0	294	0	0.78062	0	14.86	5.2771
Mg	36	1.8336	3.5246	3.12136	0.0036	8	0.0113	167	0.0068	0.86892	0.0003	3.0262	0.6410
Mn	36	2.0652	3.1335	3.95437	0.0004	10.5	0.0005	218	0.0001	0.82075	0	3.1254	1.0049
Na	36	-2.119	7.9606	-1.59746	0.1192	-4	0.2153	-54	0.3203	0.78222	0	0.574	-4.811
Zn	30	-0.002	0.0194	-0.679	0.5025	-5	0.0987	-65.5	0.1822	0.94088	0.113	0.0048	-0.0096

Legend:

Mean = Sample mean	SIGNRANK = WSR statistic
STD = Sample standard deviation	PROBS = SIGNRANK significance value
T = t statistic	NORM = Test statistic for normality
PROBT = Significance value of T	PR_N = Significance value of NORM
MSIGN = Centered sign test statistic	PROBM = Significance value of MSIGN
UCL 975 = Upper limit of the 95% confidence interval for the mean difference	
LCL 975 = Lower limit of the 95% confidence interval for the mean difference	
WSR = Wilcoxon signed rank	

methodology produced representative results. VOCs, the contaminants of concern at the KCP do not sorb onto aquifer materials, therefore, influences caused by the mobilization of colloids were expected to be minimal. Additionally, utilizing micropurge sampling techniques assumes that no mixing of water occurs between the screened portion of the well bore and groundwater lying stagnant in the well casing. Indeed, results of the statistical comparison between traditionally collected VOC samples and samples collected by the micropurge technique do not show statistically significant variations (Table 1).

Results from the comparison of metals data collected by each methodology were not expected to be comparable. It has been previously established by numerous authors (Puls and Powell, 1992, Kearn et al., 1992) that purging the well at high rates mobilizes colloidal particles, transporting otherwise immobile inorganic species. For comparison purposes, however, filtered and unfiltered metal's samples collected by traditional methods and micropurged methods were statistically compared.

A test of normality of the differences across all wells and dates was performed to determine the applicability of parametric or nonparametric tests. The tests for normality conclude that the data for both inorganic and organic compounds are not normal (at the five percent level of significance). Therefore, a nonparametric test, the Wilcoxon signed rank (WSR) test was applied. Monitor wells identified for this study were chosen with relatively low, medium, and high concentrations of VOCs, thus biasing the distribution of data non-normally.

A minimum of six valid data pairs were required for statistical comparison. Tables 1 and 2 summarize results of statistical comparisons of organic and inorganic data. Table 1 lists the summary statistics for organic compounds. Two columns are provided for each organic compound on Table 1, the first is the ranked differences of the raw data and the second is the rank. If the smallest sum of positive ranks less $n(n+1)/4$ is within the upper and lower critical value, the result is not rejected (i.e., no statistical difference). This is determined on Table 1 by comparison of the absolute value of the WSR(.01) value to the WSR calc. value. No statistical difference is determined if the WSR(.01) value is greater than the WSR calc. value. Results of the study determined no significant difference between analysis of samples collected by micropurge versus traditional methodologies for VOCs.

Table 2 lists summary statistics for comparison of inorganic constituents collected by micropurging and those collected "traditionally" and filtered with a 0.45 μ m filter. The column of most interest on this table is PROBS. Values greater than 0.05 are not significant at the five percent level meaning comparison of values determined by analysis of samples collected by the two methodologies are not statistically different. Sodium and zinc were not significant at the five percent level. Comparison of results for aluminum, arsenic, barium, calcium, iron, manganese, magnesium, and zinc were determined statistically different. As stated above, these results are not unusual. For example, it is difficult to obtain accurate values for trace levels of iron in groundwater (Urasa and Mavura, 1992). The difficulties are due to the complexity of iron's aqueous chemistry, which is greatly affected by redox conditions. The principal problem

is the sampling and analysis of colloidal iron (Kennedy et al. 1974). Neither samples collected by traditional or micropurge methodology were consistently higher for inorganic analytes.

BENEFITS

Dedicated Systems

Dedicated systems eliminate the need to collect samples to assess the effectiveness of decontamination efforts. With conventional sampling equipment, a sample blank is collected every 10 samples. With dedicated samples, the potential for cross contamination is eliminated.

Monitoring wells do not have to be re-sampled to prove that the analytical results are not affected by ineffective decontamination.

Waste Minimization

As a part of waste minimization efforts at the KCP, the process of sampling groundwater was characterized and evaluated with alternatives that would minimize the generation of both hazardous and non hazardous waste. As a result of implementing the micropurge sampling technique in 60 well completions a savings of almost 21,000 pounds of waste and \$7200.00 in waste disposal costs are realized on an annual basis.

Sampling Time

Sampling times were evaluated as a part of the project. Sampling times for wells sampled for VOCs were reduced by 75% while times for wells sampled for both metals and VOCs were reduced by 50% . Sample costs were estimated by an outside vendor to be 50% cheaper using this method.

CONCLUSIONS

The micropurge sampling methodology was found to be a more time efficient, cheaper, and, in the case of organics analysis, statistically similar to the traditional bailer method of sampling. It was also found to produce less waste than the traditional method. Samples for the analysis of metals should also be collected by micropurge sampling methodology without filtration. Numerous researchers, as referenced in this paper, have noted that sampling without filtration at low flow rates produces samples most representative of aquifer conditions.

Based on the study outlined in this paper, the KCP solicited Missouri Department of Natural Resources and EPA approval to sample groundwater utilizing the micropurge process. Regulatory approval has been obtained and dedicated sample systems will be installed in 1995 to facilitate the micropurge sampling process.

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April 15, 2009

Mr. Harold O'Connell
Ohio Environmental Protection Agency, Southwest District Office
401 East Fifth Street
Dayton, Ohio 45402

**RE: Generator Closure Requirements
Clean Ohio Restoration Fund/Voluntary Action Program Site
Former Celotex Facility
Lockland, Ohio.**

Dear Mr. O'Connell:

This letter is provided in response to our telephone conversation of April 9, 2009. The conversation was regarding my letter dated February 27, 2008, concerning Generator Closure Requirements at the referenced site. While demonstrating that Generator Closure Requirements had been met, I had used Ohio Voluntary Action Plan (Ohio VAP) industrial cleanup standards rather than Ohio Resource Conservation and Recovery Act (Ohio RCRA) Generic Cleanup Numbers (GCNs). At your request, the closure sampling results were compared to the GCNs. As expected, arsenic and benzo(a)pyrene exceeded the residential based GCNs (Table 1). Background levels of arsenic typically exceed the GCNs. The benzo(a)pyrene and other polynuclear aromatic hydrocarbons (PAHs) were detected in only one sample, a surface sample collected from beneath an area of broken asphalt pavement. These results do not appear to be due to mismanagement of the waste.

As discussed in the letter of February 27, 2008, during performance of the Cessation of Regulated Operations activities, Certainteed shipped accumulated hazardous waste off-site for disposal. The wastes included sulfuric acid, flammable materials believed to be paint thinners, and paints that were stored at the loading dock. Documentation of the accumulated waste storage area was provided in Attachment 2 of the 2008 letter. Attachment 1 of this letter provides the manifest for that waste indicating the paint thinners were D001 listed wastes.

Based on the waste types, it would be expected that volatile organic compounds (VOCs) would be present in the samples if the wastes had been mismanaged. However, no VOCs were detected in any of the samples. RCRA metals were detected in all samples collected as would be expected as all are naturally occurring. Some concentrations of the metals exceeded naturally occurring levels but considering the 100-year industrial history of the property, the levels are not unexpected. These metals would not be expected from waste mismanagement. Neither can the PAHs, which are attributed to the broken asphalt pavement.

While the results indicate the site meets the Generator Closure Requirements, a risk assessment was conducted to ensure the levels detected in the closure samples are protective of human health and the environment. The risk assessment was performed using the reasonably-expected land use. The reasonably expected land use for this property is industrial as was documented in the Ohio VAP No Further Action Letter submitted earlier this year. As such, Industrial Cleanup Numbers (ICNs) were developed following the *Closure Plan Review Guidance*. One deviation from this plan is that chemicals-of-concern (COCs) were not screened or eliminated from assessment per Section 6 of the guidance. Ohio RCRA default exposure factors were used for all calculations and were obtained from Tables A-2 and Table A-5 of the guidance.

Intake factors for ingestion, dermal contact, and inhalation were calculated (Tables 2 through 4). In Table 5, the ICNs were calculated for cancer and non-cancer risks based on slope factors and reference doses obtained from Table A-5 and Target Cancer Risk (TR) of 1E-05 and Target Hazard Index of 1.0 per Table A-1 and Section 8.0 of the guidance.

The maximum concentrations detected in the generator closure samples are compared to the ICNs in Table 6. The maximum concentrations were used since PAHs were only detected in one sample. None of the sample results exceed the ICNs and the cumulative cancer risk is less than 1.0E-5 and the cumulative hazard index is less than 1.0.

The soil ICNs are appropriate for the following reasons:

- There are no other exposure pathways at the unit in addition to those used to develop the ICNs,
- There are not large areas of contamination,
- There are no high fugitive dust levels,
- There is no cultivation of food crops, and
- There are no constituents of concern present that are not listed in Tables A-3, A-4 and A-5 of the *Closure Plan Review Guidance*.

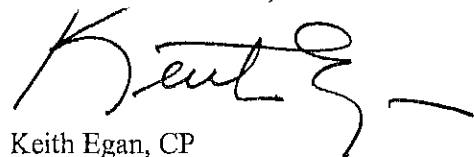
Lead was not included in the cumulative risk assessment as its risk-based GCN is not derived from a non-carcinogenic or carcinogenic endpoint. Lead is compared to its GCN and if the maximum or 95% UCL does not exceed the GCN, further evaluation is not required. The maximum and 95% upper confidence level (95% UCL) concentrations for lead are 240 and 182 mg/Kg, respectively. The residential GCN is 245 mg/Kg and as such, lead does not pose a risk to current industrial or future receptors at the property.

Sampling of the area has not demonstrated any indication that the waste accumulation at the loading dock caused impact to the environment. As such, the Generator Closure should be considered complete. Any other impact at the property has been addressed under the Voluntary Action Program.

If you have any questions concerning the closure, please feel free to call me at (513) 562-8935.

Sincerely,

Petro Environmental, LLC



Keith Egan, CP
Senior Project Manager

pc: Ihor Melnyk, CP
Joe Smindak, OEPA

TABLES

Table 1

Sample Number	Fluoranthene	Pyrene	Benzo(a)anthracene	Chrysene	Benzo(b)fluoranthene	Benzo(k)fluoranthene	Benzo(a)pyrene	Indeno(1,2,3-cd)pyrene	Benzo(g,h,i)perylene	Arsenic	Barium	Cadmium	Chromium	Lead	Mercury
LD-2	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.15	<0.33	29	200	3.4	130	240	1.1
LD-4	0.58	0.53	0.48	0.54	0.93	0.38	0.65	0.44	0.48	5.8	61	<1.0	14	20	<0.3
LD-6	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.15	<0.33	5.5	23	<1.0	4.7	4.6	<0.3
LD-8	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.33	<0.15	<0.33	12	160	<1.0	26	21	<0.3
Maximum Concentration	0.58	0.53	0.48	0.54	0.93	0.38	0.65	0.44	0.48	29	200	3.4	130	240	1.1
95% UCL	0.47	0.44	0.40	0.44	0.73	0.32	0.52	0.35	0.40	23.9	192	2.6	101	182	0.85
RCRA Direct Contact GCN	1900	1420	512	512	512	51.2	0.51	5.12	None	3.87	5170	63.5	210	245	6.68

Results in mg/Kg

Only detected chemicals are provided. There were no detections of volatile organic compounds

< - Not detected at listed detection limit

Detected results shown in **Bold**

Shaded results exceed standard

Table 2
Ingestion Intake Factor Calculation

Ingestion Intake - Cancer

Chemical of Concern	Intake Factor	EDi	IRSi	CFs	FI	EFi	BWa	ATa
Fluoranthene	1.747E-07	25	50	0.000001	1	250	70	25550
Pyrene	1.747E-07	25	50	0.000001	1	250	70	25550
Benzo(a)anthracene	1.747E-07	25	50	0.000001	1	250	70	25550
Chrysene	1.747E-07	25	50	0.000001	1	250	70	25550
Benzo(b)fluoranthene	1.747E-07	25	50	0.000001	1	250	70	25550
Benzo(k)fluoranthene	1.747E-07	25	50	0.000001	1	250	70	25550
Benzo(a)pyrene	1.747E-07	25	50	0.000001	1	250	70	25550
Indeno(1,2,3-cd)pyrene	1.747E-07	25	50	0.000001	1	250	70	25550
Benzo(g,h,i)perylene	1.747E-07	25	50	0.000001	1	250	70	25550
Arsenic	1.747E-07	25	50	0.000001	1	250	70	25550
Barium	1.747E-07	25	50	0.000001	1	250	70	25550
Cadmium	1.747E-07	25	50	0.000001	1	250	70	25550
Chromium	1.747E-07	25	50	0.000001	1	250	70	25550
Mercury	1.747E-07	25	50	0.000001	1	250	70	25550

Ingestion Intake - NonCancer

Chemical of Concern	Intake Factor	EDi	IRSi	CFs	FI	EFi	BWa	ATa
Fluoranthene	4.892E-07	25	50	0.000001	1	250	70	9125
Pyrene	4.892E-07	25	50	0.000001	1	250	70	9125
Benzo(a)anthracene	4.892E-07	25	50	0.000001	1	250	70	9125
Chrysene	4.892E-07	25	50	0.000001	1	250	70	9125
Benzo(b)fluoranthene	4.892E-07	25	50	0.000001	1	250	70	9125
Benzo(k)fluoranthene	4.892E-07	25	50	0.000001	1	250	70	9125
Benzo(a)pyrene	4.892E-07	25	50	0.000001	1	250	70	9125
Indeno(1,2,3-cd)pyrene	4.892E-07	25	50	0.000001	1	250	70	9125
Benzo(g,h,i)perylene	4.892E-07	25	50	0.000001	1	250	70	9125
Arsenic	4.892E-07	25	50	0.000001	1	250	70	9125
Barium	4.892E-07	25	50	0.000001	1	250	70	9125
Cadmium	4.892E-07	25	50	0.000001	1	250	70	9125
Chromium	4.892E-07	25	50	0.000001	1	250	70	9125
Mercury	4.892E-07	25	50	0.000001	1	250	70	9125

$$\text{Intake Factor} = \frac{\text{EDi} \times \text{IRSi} \times \text{CFs} \times \text{FI} \times \text{EFi}}{\text{BWa} \times \text{ATa}}$$

Table 3
Dermal Contact Intake Factor Calculation

Dermal Contact - Cancer

Chemical of Concern	Intake Factor	CFs	SASi	AFa	ABS	EFi	EDi	BWa	ATa
Fluoranthene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Pyrene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Benzo(a)anthracene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Chrysene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Benzo(b)fluoranthene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Benzo(k)fluoranthene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Benzo(a)pyrene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Indeno(1,2,3-cd)pyrene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Benzo(g,h,i)perylene	1.051E-07	0.000001	3300	0.07	0.13	250	25	70	25500
Arsenic	2.426E-08	0.000001	3300	0.07	0.03	250	25	70	25500
Barium	8.088E-10	0.000001	3300	0.07	0.001	250	25	70	25500
Cadmium	8.088E-10	0.000001	3300	0.07	0.001	250	25	70	25500
Chromium	8.088E-10	0.000001	3300	0.07	0.001	250	25	70	25500
Mercury	8.088E-10	0.000001	3300	0.07	0.001	250	25	70	25500

Dermal Contact - NonCancer

Chemical of Concern	Intake Factor	CFs	SASi	AFa	ABS	EFi	EDi	BWa	ATa
Fluoranthene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Pyrene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Benzo(a)anthracene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Chrysene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Benzo(b)fluoranthene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Benzo(k)fluoranthene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Benzo(a)pyrene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Indeno(1,2,3-cd)pyrene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Benzo(g,h,i)perylene	2.938E-07	0.000001	3300	0.07	0.13	250	25	70	9125
Arsenic	6.781E-08	0.000001	3300	0.07	0.03	250	25	70	9125
Barium	2.260E-09	0.000001	3300	0.07	0.001	250	25	70	9125
Cadmium	2.260E-09	0.000001	3300	0.07	0.001	250	25	70	9125
Chromium	2.260E-09	0.000001	3300	0.07	0.001	250	25	70	9125
Mercury	2.260E-09	0.000001	3300	0.07	0.001	250	25	70	9125

$$\text{Intake Factor} = \frac{\text{CFs} \times \text{SASi} \times \text{AFa} \times \text{ABS} \times \text{EFi} \times \text{EDi}}{\text{BWa} \times \text{ATa}}$$

Table 4
Inhalation Intake Factor Calculation

Inhalation - Cancer

Chemical of Concern	Intake Factor	PEF	VF	IRAI	EFi	EDi	BW _a	ATc
Fluoranthene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Pyrene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Benzo(a)anthracene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Chrysene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Benzo(b)fluoranthene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Benzo(k)fluoranthene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Benzo(a)pyrene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Indeno(1,2,3-cd)pyrene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Benzo(g,h,i)perylene	1.802E-11	1.36E+09	None	7	250	25	70	25500
Arsenic	1.802E-11	1.36E+09	None	7	250	25	70	25500
Barium	1.802E-11	1.36E+09	None	7	250	25	70	25500
Cadmium	1.802E-11	1.36E+09	None	7	250	25	70	25500
Chromium	1.802E-11	1.36E+09	None	7	250	25	70	25500
Mercury	1.802E-11	1.36E+09	None	7	250	25	70	25500

Inhalation - NonCancer

Chemical of Concern	Intake Factor	PEF	VF	IRAI	EFi	EDi	BW _a	ATc
Fluoranthene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Pyrene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Benzo(a)anthracene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Chrysene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Benzo(b)fluoranthene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Benzo(k)fluoranthene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Benzo(a)pyrene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Indeno(1,2,3-cd)pyrene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Benzo(g,h,i)perylene	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Arsenic	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Barium	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Cadmium	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Chromium	2.374E-08	1.36E+09	None	3300	250	25	70	9125
Mercury	2.374E-08	1.36E+09	None	3300	250	25	70	9125

$$\text{Intake Factor} = \frac{\text{IRAI} \times \text{EFi} \times \text{EDi} \times (1/\text{PEF}) + 1/\text{VF}}{\text{BW}_a \times \text{AT}_a}$$

Table 5
Industrial Cleanup Standards Calculation

Industrial Standard - Cancer

Chemical of Concern	Ingestion Intake (mg/kg-d)	Absorbed Dose (mg/kg-d)	Inhalation Intake (mg/kg-d)	O _{ABS}	SFo	SFi	Industrial Cleanup Number (mg/Kg)
Fluoranthene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	None	None	None
Pyrene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	None	None	None
Benzo(a)anthracene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	0.73	0.73	38.5
Chrysene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	0.0073	0.0073	3848
Benzo(b)fluoranthene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	0.73	0.73	38.5
Benzo(k)fluoranthene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	0.073	0.073	385
Benzo(a)pyrene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	7.3	7.3	38.5
Indeno(1,2,3-cd)pyrene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	0.73	0.73	38.5
Benzo(g,h,i)perylene	1.74727E-07	1.05147E-07	1.80219E-11	0.58	None	None	None
Arsenic	1.74727E-07	2.42647E-08	1.80219E-11	0.2	1.5	15	38.2
Barium	1.74727E-07	8.08824E-10	1.80219E-11	0.2	None	None	None
Cadmium	1.74727E-07	8.08824E-10	1.80219E-11	0.2	None	6.3	88076
Chromium	1.74727E-07	8.08824E-10	1.80219E-11	0.2	None	None	None
Mercury	1.74727E-07	8.08824E-10	1.80219E-11	0.2	None	None	None

Industrial Standard - NonCancer

Chemical of Concern	Ingestion Intake (mg/kg-d)	Absorbed Dose (mg/kg-d)	Inhalation Intake (mg/kg-d)	O _{ABS}	RfDo	RfDi	Industrial Cleanup Number (mg/Kg)
Fluoranthene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	0.04	None	40167
Pyrene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	0.03	None	30125
Benzo(a)anthracene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Chrysene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Benzo(b)fluoranthene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Benzo(k)fluoranthene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Benzo(a)pyrene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Indeno(1,2,3-cd)pyrene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Benzo(g,h,i)perylene	4.89237E-07	2.93836E-07	2.37424E-08	0.58	None	None	None
Arsenic	4.89237E-07	6.78082E-08	2.37424E-08	0.2	0.0003	None	362
Barium	4.89237E-07	2.26027E-09	2.37424E-08	0.2	0.07	0.00014	5658
Cadmium	4.89237E-07	2.26027E-09	2.37424E-08	0.2	0.0005	0.00005	678
Chromium	4.89237E-07	2.26027E-09	2.37424E-08	0.2	1.5	1.5	2861064
Mercury	4.89237E-07	2.26027E-09	2.37424E-08	0.2	0.000086	0.000086	164

Cancer Cleanup Number = $0.00001 / ((\text{Ingestion Intake} * \text{Sfo}) + (\text{Absorbed Dose} * \text{SFo} / \text{O}_{\text{ABS}}) + (\text{Inhalation Intake} * \text{SFi}))$

Non Cancer Cleanup Number = $1 / ((\text{Ingestion Intake} / \text{RfDi}) + (\text{Absorbed Dose} / \text{RfDo} * \text{O}_{\text{ABS}}) + (\text{Inhalation Intake} / \text{RfDi}))$

Table 6
Summary of Risk

Chemical of Concern	Industrial Cleanup Number (mg/Kg)		Closure Unit Concentration (mg/kg)	Cancer Risk	Non Cancer Hazard
	Cancer	Non Cancer			
Fluoranthene	None	40167	0.58	None	0.000014
Pyrene	None	30125	0.53	None	0.000018
Benzo(a)anthracene	38.5	None	0.48	1.25E-07	None
Chrysene	3848	None	0.54	1.40E-09	None
Benzo(b)fluoranthene	38.5	None	0.93	2.42E-07	None
Benzo(k)fluoranthene	385	None	0.38	9.88E-09	None
Benzo(a)pyrene	3.85	None	0.65	1.69E-06	None
Indeno(1,2,3-cd)pyrene	38.5	None	0.44	1.14E-07	None
Benzo(g,h,i)perylene	None	None	0.48	None	None
Arsenic	38.2	362	29	7.60E-06	0.080
Barium	None	5658	200	None	0.035
Cadmium	88076	678	3.4	3.86E-10	0.005
Chromium	None	2861064	130	None	0.000045
Mercury	None	164	1.1	None	0.007
Total				9.78E-06	0.12
Allowable				1.00E-05	1

ATTACHMENT 1

Emergency Contact Telephone Number

**Miller
Valentine
Group**

RM HAZARDOUS WASTE MANIFEST
Generator's US EPA ID No.
330 S. WAYNE ST
UNKNOWN

Manifest
Document No.
17008

2. Page 1
of 1

Information in the shaded areas is not required by Federal law.

Generator's Name and Mailing Address
330 S. WAYNE ST
DAYTON, OH 45427
Phone (513) 774-8400
Fax (513) 883-6185
Website: www.mvg.com

5. Transporter 1 Company Name
PERMA-FIX OF DAYTON, INC.
6. US EPA ID Number
OHD004274031

7. Transporter 2 Company Name
8. US EPA ID Number

9. Designated Facility Name and Site Address
PERMA-FIX OF DAYTON, INC.
300 SOUTH WEST END AVENUE
DAYTON, OH 45427
10. US EPA ID Number
OHD004274031

11. US DOT Description (Including Proper Shipping Name, Hazard Class, and ID Number)

12. Containers
No. Type

13. Total
Quantity

14. Unit
Wt/Vol

HM	No.	Type	Total Quantity	Unit Wt/Vol
a. WASTE SULFURIC ACID, 8, UN2786, PGII	44/1	DM	00250	G
b. RO WASTE FLAMMABLE LIQUID, N.O.S., 3, UN1883, PGII	003	DM	00165	G
c. RO WASTE PAINT RELATED MATERIAL, 3, UN1203, PGII	007	DF	00050	G
d. RERA EMULSION, 3, UN1883, PGII	001	DF	00100	P

Additional Description for Valuable Materials
Site Address
EMERGENCY CONTACT: CHEMTREC 1-800-424-9300

15. Special Handling Instructions and Additional Information
Site Address
EMERGENCY CONTACT: CHEMTREC 1-800-424-9300

16. GENERATOR'S CERTIFICATION: I hereby declare that the contents of this consignment are fully and accurately described above by proper shipping name and are classified, packed, marked, and labeled, and are in all respects in proper condition for transport by highway according to applicable international and national governmental regulations.

Printed/Typed Name
Signature
Month Day Year

17. Transporter 1 Acknowledgement of Receipt of Materials
Printed/Typed Name
Signature
Month Day Year

18. Transporter 2 Acknowledgement of Receipt of Materials
Printed/Typed Name
Signature
Month Day Year

19. Discrepancy Indication Space

20. Facility Owner or Operator: Certification of receipt of hazardous materials covered by this manifest except as noted in Item 19.

Printed/Typed Name
Signature
Month Day Year



Miller-Valentine Group
9435 WaterStone Blvd.
Suite 300
Cincinnati, Ohio 45249
513-774-8400
513-683-6165 Fax
Website: www.mvg.com

LOTEX CORPORATION

CINCINNATI PLANT

MOVE ORDER

#

MATERIAL DESCRIPTION:

- Hazardous Waste Drums:*
- 1) Sulfuric Acid, 250 gallons
 - 2) Misc Flammable Liquid, 165 gallons
 - 3) Inks/Paints, 50 gallons
 - 4) 5 empty 55 gallon sulfuric acid drums

REASON FOR MOVE ORDER:

- ☐ WORK TO BE DONE ☐ RETURN FOR CREDIT ☐ DOES NOT MEET SPECS
☐ PO # ☐ UNDER WARRANTY ☒ OTHER *HAZ Waste Removal*

LOCATED AT:

- ☒ SHIPPING DOCK ☐ MAINTENANCE DEPT. ☐ OFFICE
☐ OTHER (EXPLAIN)

DELIVER TO:

see attached manifest/documentation
SES, Inc & Perma Fix

ROUTE:

- ☐ VENDOR TRUCK ☒ COMMERCIAL TRUCK
☐ FREIGHT-PREPAID ☐ FREIGHT COLLECT
☐ OTHER (EXPLAIN)

(NAME)

DATE MOVED:

4/4/01

CHARGE TO:

Cincinnati Plant

APPROVALS:

[Signature]
[Signature]

REQUESTED BY *NAIOP National Plant Manager*

PLANT CONTROLLER



Miller-Valentine Group
9435 WaterStone Blvd.
Suite 300
Dayton, Ohio 45249

PERMA-FIX

ENVIRONMENTAL SERVICES

Shipper:

513-774-8400
613-663-6135 - Fixed Corp.
Website: www.mvg.com

Site Address:

330 S. Wayne Street, Lockland, Ohio 45216

Site Contact:

Derck Boyles

Phone Number:

513-874-8355

Amount Of Material:

10-Drum

Schedule Date And Time:

4/8/01 - 8:00 AM

Bill To Address:

Directions To Site:

Leave PFD	Arrive Site	Start Loading	End Loading	Leave Site	Arrive PFD
7:30 AM	8:35 AM	9:00 AM	9:30 AM	9:45 AM	

Total Miles:

116

Power Unit #:

274

Tanker/Box #:

274

Driver Comments:

Delivery Information:

Specifications:

Type Of Unit:

Vac

Pump-Tanker

☒ Box

Route Truck

Turbo

Liftgate Required:

☒ Yes

☐ No

Length Of Hose:

Fittings:

Special Instruction:

All Dock Work

Print Name:

Derck Boyles

Agent Signature: